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Postoperative care in ICU versus non-ICU after head and neck free flap surgery: A systematic review and meta-analysis

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Postoperative care in ICU versus non-ICU after head and neck free flap surgery: A systematic review and meta-analysis

Short title: ICU versus non-ICU after free flap surgery

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Abstract

Objective Admission to the intensive care unit (ICU) has long been considered as routine by most head and neck surgeons after microvascular free flap transfer. This study aimed to answer the question "Is there a difference in the flap survival and postoperative complications rates between admission to intensive care unit (ICU) versus Non-ICU following microvascular head and neck reconstructive surgery?".

Design Systematic review, and meta-analysis.

Methods the PubMed, Embase, Scopus, and Cochrane Library electronic databases were systematically searched (till April 2020) to identify the relevant studies. Studies that compared postoperative nursing of patients who underwent microvascular head and neck reconstructive surgery in ICU and non-ICU were included. The outcome variables were flap failure and length of hospital stay (LOS) and other complications. Weighted Odd ratio or mean differences with 95% confidence intervals were calculated.

Results Eight studies involving a total of 2349 patients were included. No statistically significant differences were observed between ICU and non-ICU admitted patients regarding flap survival reported (Fixed, RR, 1.49; 95% CI, 0.820; 2.727, P = 0.190, I^2 = 0%), reoperation, readmission, respiratory failure, delirium, and mortality (P>0.05). A significant increase in the postoperative pneumonia (P=0.018) and sepsis (P=0.033) was observed in patients admitted to ICU compared to non-ICU setting.

Conclusion This meta-analysis showed that an immediate postoperative nursing in the ICU after head and neck microvascular reconstructive surgery did not reduce the incidence of flap failure or complications rate. Limiting the routine ICU admission to the carefully selected patients may result in a reduction in the incidence of postoperative pneumonia, sepsis, LOS and total hospital charge.

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Keywords**:** *Microvascular free flap; intensive care unit; flap survival; complications; postoperative care.*

Strengths and limitations of this study

- *This is the first systematic review and meta-analysis demonstrating that patient who underwent head and neck microvascular reconstructive surgery may not require routine postoperative care in ICU.*
- *No difference was observed between ICU and Non-ICU care concerning flap survival rate, reoperation, readmission, respiratory failure, delirium, and mortality.*
- *We analyzed data from both randomized trials and observational studies*
- *Due to heterogeneity in the reported data, meta-analysis concerning total hospital cost was not performed.*

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Introduction

Recently, microsurgical procedures including Free flap reconstruction have gained a wide popularity becoming the standard of care for various surgical defects of the head and neck region [1]. Free flap surgery with various available flaps having different characteristics covers a wide range of indications in reconstructive head and neck surgery with different perspectives and an overall success rate reported up to 95% [2].

Being considered as a major surgery with long intraoperative time, the postoperative care of free flap surgery is considered critical and requires great attention. In addition, the need to closely monitor the flap viability for any vascular compromise occurring mostly in the first 48 to 72 hours,

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increases the need for an accurate and close monitoring in the immediate post-operative interval, therefore this interval is critical and of utmost importance for successful salvage of ischemic flaps [3]. Nevertheless, still which postoperative protocol is the most suitable is an issue of controversy and there is no international consensus between surgeons regarding a common monitoring regimen [4].

For years, admission to the intensive care unit (ICU) has long been considered as routine by most head and neck reconstructive surgeons following free flap surgery with a mean of 2.4 days admission in the ICU [5]. It was reported that 88.9% of surgeons prefer sending their free flap patients to ICU even though most patients does not meet the cardiopulmonary indications of ICU admission [6][7]. In historic literature, certain post-operative treatment regimens stated complete patient immobilization during the immediate post-operative period in an effort to protect the vascular pedicle [8]. This has evolved to the most common accepted standard where patients are spontaneously breathing at the conclusion of surgery. However, admission to the ICU is not without drawbacks due to prolonged mechanical ventilation, late ambulation, overuse of sedation, in addition to the higher incidence of pneumonia [8][9]. Accordingly, a different era of postoperative care of free flap patients has been evolved recently in practice entailing the postoperative care at ward level or non-ICU postoperative care, with almost no significant differences in morbidity, mortality or flap failure between the ICU and ward level postoperative care, but with lengthy hospital stay in those patients admitted to ICU [10][11]. Reviewing the literature, there is no clear consensus which post-operative free flap management protocol is better; an ICU admission versus non-ICU care. Due to lack of the evidence regarding which postoperative free flap nursing is associated with the highest flap survival and lower complications rates, therefore, the current systematic review and meta-analysis aimed to

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summarize the available evidence and to answer the question "Is there a difference in the flap failure and postoperative complication rates between the patients who postoperatively admitted to intensive care unit (ICU) or Non-ICU after microvascular head and neck reconstructive surgery?"

Methods

This systematic review and meta-analysis adhered to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement for reporting systematic reviews [12] (Supplemental Table S1). The protocol of the meta-analysis has been registered on the PROSPERO platform (CRD42020207772).

Focused question

Is there a difference in the flap survival and postoperative complications rates between admission to intensive care unit (ICU) versus Non-ICU following microvascular head and neck reconstructive surgery?

The question for the current meta-analysis was adopted to follow PICO criteria:

P: Patients who underwent tumor resection and free flap reconstruction.

I: Participants who received nursing in ICU or non-ICU department.

C: ICU versus non-ICU admission

O: flap survival rate reoperation, readmission, and respiratory complications, delirium, sepsis, Length of hospital stays, or Hospital cost.

Search strategies

From inception to April 2021, an electronic search without time or language restrictions was done in the following databases: PubMed, Embase, Scopus, and the Cochrane library (Supplemental Table S2). The following terms were used in the search strategy on PubMed:

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(((((((head and neck neoplasms [MeSH Terms]) OR (microsurgical free flap [MeSH Terms])) AND (nursing, perioperative [MeSH Terms])) OR (specialized care unit)) AND (ward care)) OR (care, surgical intensive [MeSH Terms])) AND (postoperative complications [Title/Abstract])) OR (flap failure[Title/Abstract]).

A manual search of the following journals was also performed: International Journal of surgery, Journal of plastic and reconstructive surgery, Journal of plastic, esthetic and reconstructive surgery, Journal of annals of plastic surgery, Journal of Intensive care, Journal of Intensive care medicine, Journal of Intensive care, Journal of head and neck, Laryngoscope, British Journal of Oral and Maxillofacial Surgery, International Journal of Oral and Maxillofacial Surgery, Journal of Craniofacial Surgery, Journal of Cranio-Maxillofacial Surgery, Journal of Microsurgery, Oral Surgery Oral Medicine Oral Pathology Oral Radiology and Endodontology, Journal of head and neck surgery and JAMA otolaryngology head and neck surgery. The reference list of the identified studies and the relevant reviews on the subject were also evaluated for possible additional studies. Moreover, online databases providing information about clinical trials in progress were checked (clinicaltrials.gov; www. centerwatch.com/clinical trials; www.clinicalconnection.com).

Inclusion and Exclusion Criteria

Eligibility criteria included human experimental or observational studies that compared the immediate postoperative admission of patients undergoing head and neck reconstructive surgery to ICU versus non-ICU and reported one of the following outcomes: Flap failure, reoperation, readmission, and respiratory complications, delirium, sepsis, Length of hospital stays, or Hospital cost. The exclusion criteria were other studies that reported one of the following: 1) Studies not comparing the postoperative ICU versus No-ICU care. 2) Studies with < 30 patients in each

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group. 3) Studies including an anticipated need for post-operative ICU care. 4) Studies including patients under the age of 18, 6) Review studies, meeting abstracts, and/or non-English articles.

Data extraction process

Two researchers independently assessed the titles, abstracts, and full-texts of the relevant studies and any controversy was resolved by discussion to reach a consensus. The following data were collected for each study (when available): authors, publication year, country of origin, study design, Age of the participants, number of patients in each group, Length of ICU admission, length of hospital stay (LOS), nursing ratios during study, total hospital charge, number of flap failure in each group, and other outcomes (Tables 1). Two researchers independently reviewed the included articles and collected the data. Disagreements between the reviewers were resolved by discussion between and if necessary, a third reviewer was involved to reach a consensus. A further contact with authors for possible missing data was also performed.

Patient and public involvement

The present study was a meta-analysis and systematic review based on published data. Patients and public were not involved in the study design, conduct, data analysis and result dissemination.

Methodological quality appraisal

Two authors independently assessed the risk of bias in the included studies. The quality assessment was performed by combining the proposed criteria of the Meta-Analysis of Observational Studies in Epidemiology (MOOSE) statement[13], Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement[14], and the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement[12]. The classification of the risk of bias potential for each study was based on five criteria: (1) random selection in the population, (2) definition of inclusion and exclusion criteria, (3) report of losses to follow-up, (4) validated

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measurements, and (5) statistical analysis. A study that met all the above-mentioned criteria was classified as having a low risk of bias, whereas a study that did not meet one of these criteria was classified as having a moderate risk of bias. When two or more criteria were not met, the study was considered to have a high risk of bias. (Table 2).

Statistical Analysis

For binary data, an odd ratio (OR) was estimated. For continuous data, the mean difference (MD) was estimated. The weighted OR or MD along with 95% CI were used to construct forest plots of the selected studies. The I^2 statistic were used to express the percentage of the total variation across studies due to heterogeneity; I^2 with 25% corresponding to low heterogeneity, 50% to moderate, and 75% to high[15]. When I^2>50%, the random effect model was used as described by DerSimonian and Laird [16]. Otherwise, the data were regarded as homogeneous, and a fixed-effect model was used. A p-value of <0.05 was considered statistically significant[16]. A Comprehensive Meta-Analysis Software version 3 (Biostat Inc, Englewood, NJ) was utilized for data analysis.

Results

The kappa value was 0.85, so the agreement between investigators was almost perfect. The electronic and manual searches identified 2286 articles (Fig. 1); 1754 records remained after removal of the duplicates. The titles and abstracts of the remaining 1754 articles were screened and 1734 records were excluded due to being topic off or non-English studies. The remaining 20 studies were carefully read by two researchers for potential inclusion. Of the 20 full-text studies reviewed for potential inclusion; 8 met the inclusion criteria and were assessed for reliability (Table 1), with the other 12 articles excluded with reasons (Supplemental Table S3). The 8 included articles had a total of 2349 participants. One study was a prospective randomized

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controlled trials [17] and the other studies were nonrandomized studies[9,11,18–22]. Other general characteristics of the included studies were reported in the Table 1

Table 1: Characteristics of the included studies.

Study	Study design	Age	Participants	Length of ICU admission	Nursing ratios during study	LOS	Flap Failure	Total hospital charges (mean)	outcomes
Nkenke et al. 2009 Germany	PS	ICU=62.8 (9.7) years Non-ICU= 63.5 (9.5)	ICU=50 Non-ICU=50	3.5 (10.9)	NR	mean, ICU=10.28, Non-ICU= 9.89	1	NR	Flap failure and other complications, Readmission, Length of hospital stays, Hospital cost and systemic complications
Arshad et al. 2013 USA	RS	ICU=59.2 Non-ICU=58.	ICU=131 Non-ICU=1	NR	NR	mean, ICU=10.28, Non-	9	$2134 > non-ICU	Flap failure, Length of hospital stays, Hospital cost and systemic

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		9	26			ICU= 9.89		proto col	complications.
Yang et al. 2019 USA	RS	64.28 (11.70)	ICU=205 Non-ICU=100	NR	NR	ICU=6 (5–8) 45.9; Non-ICU= 5 (5–7)	9	35% higher hospital costs in ICU	Flap failure, Reoperation, Readmission, length of hospital stays, systemic complications.
Panwar et al. 2015 USA	RS	ICU= mean 62.46(13.5) Non-ICU= 63.3 (11.6)	ICU=175 Non-ICU=72	NR	1:3 to 1:4	ICU=8 (7–11) Non-ICU= 7 (6–9.5)	9	ICU= $33,642 ($28,143–$43,196 ;Non-ICU= $28,524 ($22,611–	Flap failure, in-hospital mortality, and morbidity, hospital length of stay (LoS), total hospital-based charges, and cost of care, Operative time, ischemia time, site of tissue harvest.

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								$33,226)	
Chen et al. 2018 Taiwan	RS	ICU=53.8 ± 10.3 Non-ICU=52.2 ± 9.6	ICU=138 Non-ICU=179	NR	1:2 in the ICU; 1:9 during the day, 1:12 in the evening, and 1:16 in night shift in the ward	ICU=22.1 ± 13.1 Non-ICU=24.2 ± 13.9	8	NR	ventilation length, use of sedation, flap outcomes, flap complications, systemic complications, in-hospital mortality, and hospitalization. Flap outcomes
Yalamanchi et al. 2020 USA	RS	ICU= mean 64.6 SD (13.0) Non-ICU= mean 61.5 SD (13.2)	338 patients, ICU=146 Non-ICU= 192	Average 7 (IQR 6-9) days	1:3	ICU=8.5 days (IQR 6–9) Non-ICU=8 days (IQR 4–8)	13	ICU= mean $47,315.44 Non-ICU= $38,853.50	Postoperative complications, reoperation, and flap failure, ICU length of stay, total LOS, readmission rates, and hospital cost.

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Cervenka et al. 2019 USA	RCT	>18	111 patients, ICU=57 Non-ICU=61	ICU for 12–24 h	1:3 and 1:4	ICU= mean 8.89 (4–30). Non-ICU= 8.97 (4–21).	1	ICU= mean $466,199 Non-ICU= $486,150	LOS, flap failure rate, surgical and medical complications, and total cost of hospitalization
Clemens et al. 2015, USA	RS	60 (14.1) years	680 ICU= 605 Non-ICU= 75	NR	NR	Non-ICU= 5.9 6(1.8) ICU= 9.56 (7.5)	10	NR	Flap failure rate, LOS, other complications

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Quality assessment

One study was considered as having low risk of bias, six studies[11,18–22] were scored as moderate risk of bias and one study [9] was scored as having high risk (Table 2).

Results of individual outcome variables

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Flap survival

Flap survival was reported in 8 studies [9,11,17–22] with a total of 2349 patients. The flap survival rate was 97.19% in the ICU group and 97.89% in the non-ICU admission; however, a statistically insignificant difference was reported (Fixed, RR, 1.53; 95% CI, 0.821; 2.828, P = 0.182, I^2 = 0%) (Fig 2).

Authors/ Year published	Random selection	Defined inclusion/	Loss of follow-	Validated measurement	Statistical analysis	Estimated potential

Reoperation and re-admission to the ICU

Reoperation was reported in 5 studies [11,17,20–22] with a total of 1253 patients. The incidence of reoperation was 12.34% in ICU and 9.44% in non-ICU admissions; however, a statistically insignificant difference was reported (Random, RR, 0.830; 95% CI, 0.540; 1.276, P = 0.397, I^2 = 53%) (Table 3). Readmission to ICU was reported in four studies [17,18,20,22] with a total of 861 patients. The incidence of readmission was 10.91% in ICU and 8.95% in non-ICU admissions; however, a statistically insignificant difference was reported (Fixed, RR, 0.922; 95% CI, 0.611; 1.392, P = 0.700, I^2 = 0%) (Table 3).

Medical complications

- The incidence of pneumonia was reported in four studies [17–20] (6.87% in ICU and 2.45% in non-ICU) with a total of 1203 patients. There was a statistically significant higher incidence of pneumonia in ICU compared to non-ICU nursing (Fixed, RR, 2.769; 95% CI, 1.193; 6.424, P = 0.018, I^2 = 0%) (Fig 3).

Table 2. Results of the quality assessment.

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	in population	exclusion criteria	up			risk of bias
Nkenke et al. 2009, Germany	No	Yes	Yes	Yes	Yes	Moderate
Arshad et al. 2013, USA	No	No	Yes	Yes	Yes	High
Yang et al. 2019 USA	No	Yes	Yes	Yes	Yes	Moderate
Panwar et al. 2015, USA	No	Yes	Yes	Yes	Yes	Moderate
Chen et al. 2018, Taiwan	No	Yes	Yes	Yes	Yes	Moderate
Yalamanchi et al. 2020, USA	No	Yes	Yes	Yes	Yes	Moderate
Cervenka et al. 2019, USA	Yes	Yes	Yes	Yes	Yes	Low
Clemens et al. 2015, USA	No	Yes	Yes	Yes	Yes	Moderate

Outcome Variable	No. of study	No. of patients	Incidence rate (%)	Effect size (95% CI) Risk Ratio; Lower limit- upper limit	Heterogeneity test	P value

Table 3. Summary of the results of the individual outcomes.

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Flap survival	8	2349	ICU=97.19 in Non-ICU=97.89%	RR, 1.49; CI, 0.820; 2.727	0%	0.182
Partial flap necrosis	2	880	ICU=1.98 Non-ICU=2.4	RR, 1.251; CI,0.321; 4.876	0	0.747
Venous congestion	3	1097	ICU=2.90 Non-ICU=3.39	RR, 1.047; CI, 0.486; 2.257	0	0.906
Arterial thrombosis	3	1097	ICU=1.01 Non-ICU=1.97	RR, 0.804; CI,0.261; 2.471	0	0.703
Reoperation	5	1253	ICU= 12.34% non-ICU=9.44%	RR, 1.327; CI, 0.693; 2.539	53%	0.394
Readmission	4	861	ICU=10.91% Non-ICU=8.95%	0.922; 95% CI, 0.611; 1.392	0	0.700
Pneumonia	4	1203	ICU=6.87 Non-ICU=2.45	RR, 2.769; CI, 1.193;6.424	0	0.018
Respiratory failure	3	1027	ICU=11.33 Non-ICU=6.09	RR, 1.675; CI,0.911;3.078	35%	0.097
Need for ventilation	2	365	ICU=12.5 Non-ICU=10.52	RR, 1.926; CI, 1.128;3.288	0	0.016
Sepsis	5	1462	ICU=1.6 Non-ICU=0	RR, 4.23; CI, 1.12; 15.91	0	0.033
Delirium	2	365	ICU=12.5	RR, 2.067; CI, 0.944;	0	0.069

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			Non-ICU=6.02	44.526		
Mortality	3	1115	ICU=0.38 Non-ICU=0.32	RR, 0.633; CI, 0.072; 5.532	0	0.679

- The incidence of sepsis was reported in 5 studies [11,17–19,21] (1.6% in ICU and 0% in non-ICU) with a total of 1462 patients. There was a statistically significant higher incidence of sepsis in the ICU compared to non-ICU (Fixed, RR, 4.23; CI, 1.12; 15.91, P = 0.033, I2 = 0%) (Fig 4).
- No statistically significant difference was reported when comparing ICU and non-ICU concerning delirium, respiratory failure, and mortality (P> 0.05) (Table 3).

Length of hospital stay (LOS) and total hospital charge (THC)

LOS was reported in 4 studies [17,19,21,22] with a total of 773 patients. There was a statistically insignificant increase in LOS of about 1.5 day in the ICU group (Random, MD, 1.5; 95% CI, -0.835; 3.861, P = 0.270) (Fig 5). We were unable to conduct meta-analysis regarding hospital charge due to heterogenety and missing of some data. Cervenka et al[19], reported no significant difference between ICU and Non-ICU nursing regarding THC. Yang et al [20] stated that THC is about 35% higer in the patients admitted to ICU. Panwar et al[11] and Yalamanchi et al [22] reported a statistically significant difference between ICU and Non-ICU caring P<0.001 and 0.00054 respectively.

Discussion

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The current systematic review and meta-analysis is novel in that it provides an evidence to suggest that patients who will undergo head and neck microvascular reconstructive surgery may not require routine postoperative ICU care, and the non-ICU nursing can provide equivalent clinical outcomes. This was consistent with several reported studies [9,11,18,23,24], however, this concept does not mean that those patients do not require a special care, but, it is feasible if an appropriate environment and adequately-trained nursing staff are available [25]. Bhama et al compared two types of ICU care (open versus closed) of patients with head and neck free flap surgeries and they did not report any significant differences in the postoperative outcomes between the two nursing policies. The current meta-analysis included 8 studies with a total of 2349 patients and showed a statistically insignificant difference between ICU and non-ICU admission concerning flap survival rate. The pooled flap survival rate was 97.19 % in the ICU group and 97.89% in the non-ICU group and this was consistent with the results obtained from other studies [24,26]. This result indicates that the postoperative admission in ICU or non-ICU settings does not influence flap survival rate. This was in line with the study performed by Chen et al, in that they showed no significant difference in the flap survival rate between ICU and non-ICU caring and they reported flap success rate of 96.7% and 98.3% respectively.

The common justification for routine postoperative care in the ICU of the patients who underwent microvascular free flap reconstruction is close flap monitoring and airway management [22]. However; close flap monitoring can also be achieved in the non- ICU setting; provided that this is done by an adequately trained nursing staff and in a controlled environment, and this includes an hourly physical examination of the flap to assess color, capillary refill, tissue temperature, turgor, pinprick test and Doppler signals.

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On the other hand, to secure the airway after head and neck free flap reconstruction and simultaneously decreasing the need for postoperative ICU care, Nkenke et al[18] considered primary tracheostomy as a key point that allows avoiding the need for ICU admission. In a postal survey performed by Marsh et al.[7] in UK, they concluded that 69% of oral and maxillofacial surgeons prefer to perform tracheostomy after head and neck free flap surgery. However, tracheostomy is still not without complications (amounted of about 8-45%), [27,28] being chest infection the commonest complication (8%)[25].

The reported incidence of readmission to ICU in the present study for previously nursed patients in ICU or non-ICU setting was 10.91% and 8.95% respectively, and also no statistically significant difference was reported (P=0.7). These results indicated that patients underwent head and neck flap transfer might need readmission to ICU due to late postoperative complications, and these complications may happen regardless of the initial postoperative care in ICU or non-ICU setting.

Some surgeons belief that one of the benefits of the ICU admission is to limit patient mobilization by using sedative medications and this would keep the vascular patency by preventing kinking of microvascular anastomoses and allows more invasive hemodynamic monitoring [9], however, an evidence supporting or conflicting this belief is still lacking. In addition, admission to ICU is not free of disadvantages including nosocomial infections, cancellation of operation due to limited bed availability, and increased cost [10]. Moreover, the use of sedation may decrease systemic blood pressure and subsequent reduction in blood perfusion of the flap [4].

Postoperative pulmonary complications are one of the most frequent complications in patients undergoing radical head and neck cancer surgery with free flap reconstruction, accounted for about 32% (pulmonary edema in 23.6% and pneumonia in 9.1%)[29]. Saied et al [30] conducted a large study on 14212 patients who were admitted to the ICU for more than 48 hour. They reported the

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incidence of two types of pneumonia in patients admitted in the ICU; the ventilator-associated pneumonia 1,161(15%) and ICU–hospital-acquired pneumonia 176 (2%). The current meta-analysis showed a statistically significant higher incidence of postoperative pneumonia and sepsis in the ICU nursing group compared to Non-ICU, P= 0.018 and 0.033 respectively. The pooled incidences of pneumonia in ICU and non-ICU were 6.4% and 2.8% respectively, meaning that patients admitted to ICU are at higher risk of development of ventilator-associated pneumonia or ICU–hospital-acquired pneumonia compared to non-ICU admitted patients.

We were unable to perform a meta-analysis concerning hospital charge because of the heterogeneity in the reported data. However, according to the existing literature, ICU care fee reported to be accounted for about 35% of the total hospital cost.[20] Therefore, admission to non-ICU care could reduce the high cost of ICU services and facilitates the management of available resources.

Although this systematic review and meta-analysis is the first to give a critical summary of the literature, pointing towards the unnecessity to nurse the carefully selected patients undergoing microvascular head and neck reconstructive surgery in the ICU setting. However, some limitations of this study should be declared. First, only one RCT was included. Second, several cofounders that might have an effect on the postoperative outcomes were not evaluated in the included studies such as the level and specialist training of the staff, nurse to patient's ratio, amount of intraoperative fluid used, use of vasopressors and anticoagulant … etc. Third, although there was a consensus among the included studies in that ICU care increases the total hospital cost of care, no meta-analysis was performed due to both missing and heterogeneity in the reported data. Nevertheless, the current meta-analysis provides an evidence regarding the feasibility of nursing the carefully selected patients undergoing head and neck free flap surgery in non-ICU setting.

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Conclusion

The present systematic review and meta-analysis demonstrates that an immediate postoperative ICU admission of patients who have undergone microvascular head and neck reconstructive surgery did not reduce flap failure and postoperative complication rates. However, limiting the routine ICU admission to the carefully selected patients may result in a significant cost savings and reduction in the incidence of postoperative pneumonia and sepsis. Regardless where was the nursing of the patients who underwent head and neck microvascular flap surgery, the availability of an appropriate environment and adequately-trained nursing staff are the main key points and essential in the management of the postoperative complications.

Footnotes

- **Contributors** MAM and TA conceived and designed the study. MAM and TA undertook data extraction and quality assessments. MAM, HA and ZSY performed statistical analyses. HA, TA and WLP prepared the figures and tables. All the authors provided inputs and were involved with the interpretation of the results. MAM and AA drafted the manuscript. MAM, AA and WLP revised the manuscript. All the authors have read and approved the final version.
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- **Ethics approval:** Not applicable. This article does not contain any studies with human participants performed by any of the authors.
- **Competing interests** None declared.

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- **Data availability statement:** All data relevant to the study are included in the article or uploaded as supplementary information.

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Figures

Figure 1

Study flow diagram

Figure 2

Forest plot of flap survival rate.

Figure 3

Forest plot of the incidence of pneumonia

Figure 4

Forest plot of the incidence of sepsis

Figure 5

Forest plot of length of hospital stay

Supplemental materials

Table S1. PRISMA checklist.

Table S2. Search terms and results of Search strategy

Table S3. Excluded study and reasons for exclusion

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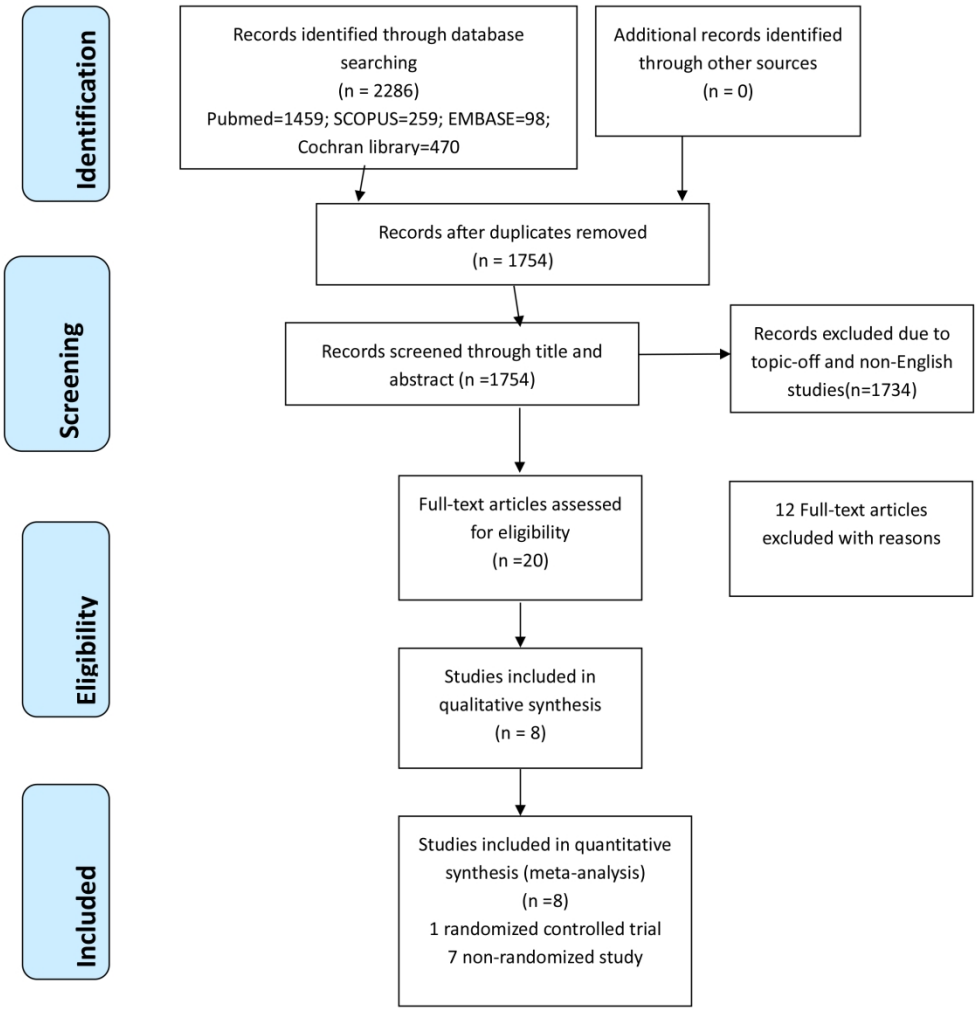


Figure 1. Study flow diagram

Figure 1
Study flow diagram

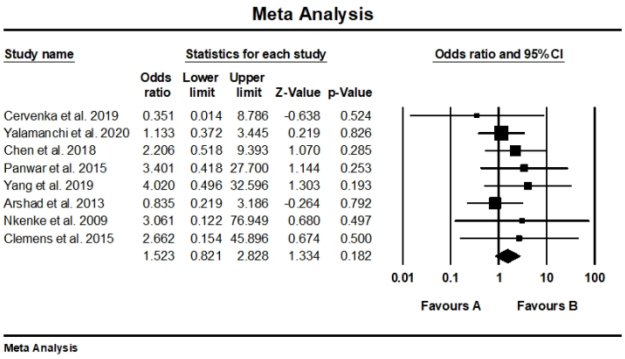


Figure 2
Forest plot of flap survival rate.

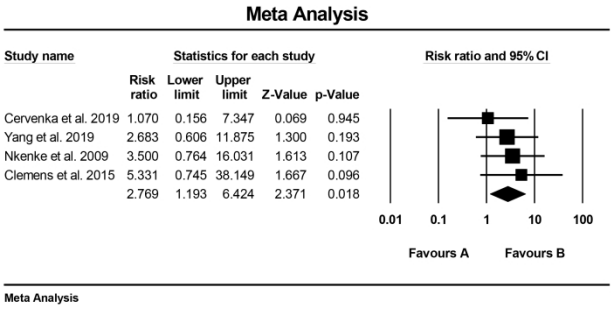


Figure 3
Forest plot of the incidence of pneumonia. There was statistically significant higher incidence of pneumonia in ICU compared to non-ICU nursing (Fixed, RR, 2.769; 95% CI, 1.193; 6.424, P = 0.018, I2 = 0%). The horizontal lines indicate the 95% CI for the 4 included studies. The squares in the middle of the lines indicate the mean effect of the study. The diamond shape at the bottom indicates the 95% CI for the pooled effect.

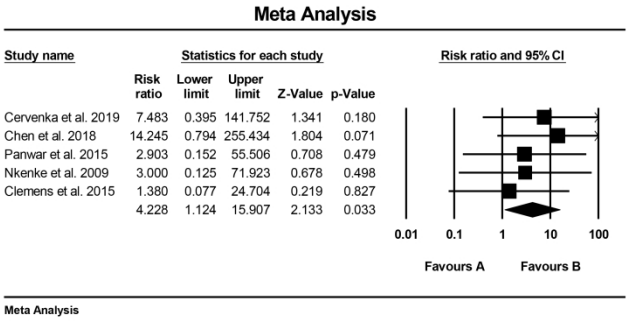


Figure 4

Forest plot of the incidence of sepsis. There was statistically significant higher incidence of sepsis in ICU compared to non-ICU nursing (Fixed, RR, 4.23; CI, 1.12; 15.91, P = 0.033, I2 = 0%). The horizontal lines indicate the 95% CI for the 5 included studies. The squares in the middle of the lines indicate the mean effect of the study. The diamond shape at the bottom indicates the 95% CI for the pooled effect.

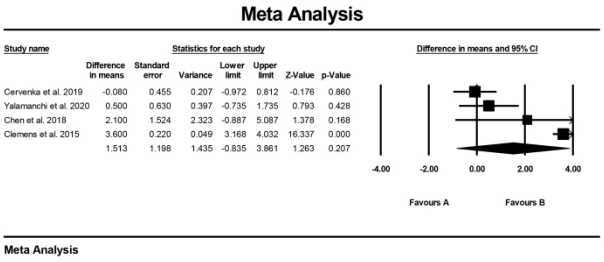


Figure 5
Forest plot of length of hospital stay (LOS). There was a statistically insignificant increase in LOS of about 1.5 day in the ICU group compared to non-ICU protocol. The horizontal lines indicate the 95% CI for the 4 included studies. The squares in the middle of the lines indicate the mean effect of the study. The diamond shape at the bottom indicates the 95% CI for the pooled effect.



PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			1,2
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	1,2
INTRODUCTION			2,3
Rationale	3	Describe the rationale for the review in the context of what is already known.	2,3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	3
METHODS			4-7
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	4
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	4
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4,5
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	5
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	5,6
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	6
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	6,7
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	6,7

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	6
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	7
RESULTS			7-9
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	7
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	7
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	7
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	8,9
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	8,9
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	-
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	-
DISCUSSION			9-12
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	9,10
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	11,12
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	10,11
FUNDING			NR
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	NR

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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Table S1: Search terms

	Search terms	No.
Pubmed 29-04-2021	(((((((head and neck neoplasms[MeSH Terms]) OR (microsurgical free flap[MeSH Terms])) AND (nursing, perioperative[MeSH Terms])) OR (specialized care unit)) AND (ward care)) OR (care, surgical intensive[MeSH Terms])) AND (postoperative complications[Title/Abstract])) OR (flap failure[Title/Abstract])	1459
SCOPUS 29-04-2021	head and neck tumor OR free flap reconstruction AND intensive care unit OR surgical intensive care unit AND intensive care unit OR surgical intensive care unit OR specialized care AND graft failure OR postoperative complications	259
EMBASE 29-04-2021	('head and neck tumor'/exp OR 'free flap reconstruction'/exp) AND ('intensive care unit'/exp OR 'surgical intensive care unit'/exp) AND ('intensive care unit'/exp OR 'surgical intensive care unit'/exp OR 'specialized care'/exp) AND ('graft failure'/exp OR 'postoperative complications'/exp)	98
Cochran Library 29-04-2021	#1 MeSH descriptor: [Head and Neck Neoplasms] explode all trees #2 specialized care unit #3 MeSH descriptor: [Free Tissue Flaps] explode all trees #4 MeSH descriptor: [Intensive Care Units] explode all trees #5 MeSH descriptor: [Postoperative Complications] explode all trees 39311 #6 flap loss #7 #1 OR #3 AND #2 AND #4 AND #6 #8 #7 AND #5	470

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Table S2 Excluded studies and reason for exclusion

	Study	Reason for exclusion
1	Godden et al. 1999 [1]	Included patients who underwent tumor resection without flap reconstruction
2	Moreno and Bonilla-Velez[2]	Compared three different protocols. Two protocol admitted to the ICU for overnight before transfer to the intermediate care and to the general floor at physician discretion.
3	EDWARD et al. 2002 [3]	Although the author stated that some patients were unadmitted to the ICU, but no outcome were reported regarding patients in Non-ICU setting
4	Vimawala et al[4]	Conference poster
5	Morse et al 2019 [5]	All groups admitted to the ICU before transfer to the Non-ICU setting.
6	Yu et al. 2019 [6]	All patients were transferred to the intermediate level care.
7	Mathew et al 2010[7]	Only 7 patients were transferred to the ICU
8	Dautremont et al 2013[8]	Patients in both groups were admitted to the ICU.
9	Bhama et al 2013[9]	Patients were admitted to open or close ICU.
10	Patel et al. 2017 [10]	No clear information about the three-protocols used (ICU, intermediate level, and surgical ward).
11	Al-Dam et al. 2014[11]	No clear information about the two-protocol used (ICU, intermediate level). Flap failure or other outcomes of interest were not reported in both groups.
12	Bertelsen et al. 2020 [12]	Two different protocols were used but, both group were admitted to the ICU for overnight before transfer to the ward.

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Primary Subject Heading:	Surgery
Secondary Subject Heading:	Dentistry and oral medicine, Ear, nose and throat/otolaryngology, Intensive care, Surgery
Keywords:	Facial plastic & reconstructive surgery < OTOLARYNGOLOGY, SURGERY, Head & neck surgery < OTOLARYNGOLOGY

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Postoperative care in ICU versus non-ICU after head and neck free flap surgery: A systematic review and meta-analysis

Short title: ICU versus non-ICU after free flap surgery

Mubarak Ahmed Mashrah[1, *], Taghrid Aldhohrah [2], Ahmed Abdelrehem[3], Bahia Sabri [2], Hyat Ahmed [4], Natheer H Al-Rawi[5], Tian Yu[6], Shiyong Zhao [1], Liping Wang [1, *], Linhu Ge [1, *]

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Abstract

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Objective Admission to the intensive care unit (ICU) has long been considered as routine by most head and neck surgeons after microvascular free flap transfer. This study aimed to answer the question "Is there a difference in the flap survival and postoperative complications rates between admission to intensive care unit (ICU) versus Non-ICU following microvascular head and neck reconstructive surgery?"**.**

Design Systematic review, and meta-analysis.

Methods the PubMed, Embase, Scopus, and Cochrane Library electronic databases were systematically searched (till April 2021) to identify the relevant studies. Studies that compared postoperative nursing of patients who underwent microvascular head and neck reconstructive surgery in ICU and non-ICU were included. The outcome variables were flap failure and length of hospital stay (LOS) and other complications. Weighted Odd ratio or mean differences with 95% confidence intervals were calculated.

Results Eight studies involving a total of 2349 patients were included. No statistically significant differences were observed between ICU and non-ICU admitted patients regarding flap survival reported (Fixed, RR, 1.46; 95% CI, 0.80; 2.69, P = 0.231, I^2 = 0%), reoperation, readmission, respiratory failure, delirium, and mortality (P>0.05). A significant increase in the postoperative pneumonia (P=0.018) and sepsis (P=0.033) was observed in patients admitted to ICU compared to non-ICU setting.

Conclusion This meta-analysis showed that an immediate postoperative nursing in the ICU after head and neck microvascular reconstructive surgery did not reduce the incidence of flap failure or complications rate. Limiting the routine ICU admission to the carefully selected patients may result in a reduction in the incidence of postoperative pneumonia, sepsis, LOS and total hospital charge.

Keywords**:** *Microvascular free flap; intensive care unit; flap survival; complications; postoperative care.*

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Strengths and limitations of this study

- *This is the first systematic review and meta-analysis to evaluate the postoperative outcomes of nursing in ICU versus non- ICU after head and neck microvascular reconstructive surgery.*
- *We analyzed data from both randomized trials and observational studies.*
- *For some outcomes, the numbers of included studies were few.*
- *The major limitation of this study will be the high heterogeneity Between the published studies.*

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Introduction

Recently, microsurgical procedures including Free flap reconstruction have gained a wide popularity becoming the standard of care for various surgical defects of the head and neck region [1]. Free flap surgery with various available flaps having different characteristics covers a wide range of indications in reconstructive head and neck surgery with different perspectives and an overall success rate reported up to 95% [2].

Being considered as a major surgery with long intraoperative time, the postoperative care of free flap surgery is considered critical and requires great attention. In addition, the need to closely monitor the flap viability for any vascular compromise occurring mostly in the first 48 to 72 hours, increases the need for an accurate and close monitoring in the immediate post-operative interval, therefore this interval is critical and of utmost importance for successful salvage of ischemic flaps [3]. Nevertheless, still which postoperative protocol is the most suitable is an issue of controversy and there is no international consensus between surgeons regarding a common monitoring regimen [4].

For years, admission to the intensive care unit (ICU) has long been considered as routine by most head and neck reconstructive surgeons following free flap surgery with a mean of 2.4 days admission in the ICU [5]. It was reported that 88.9% of surgeons prefer sending their free flap patients to ICU even though most patients does not meet the cardiopulmonary indications of ICU admission [6][7]. In historic literature, certain post-operative treatment regimens stated complete patient immobilization during the immediate

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post-operative period in an effort to protect the vascular pedicle [8]. This has evolved to the most common accepted standard where patients are spontaneously breathing at the conclusion of surgery. However, admission to the ICU is not without drawbacks due to prolonged mechanical ventilation, late ambulation, overuse of sedation, in addition to the higher incidence of pneumonia [8][9]. Accordingly, a different era of postoperative care of free flap patients has been evolved recently in practice entailing the postoperative care at ward level or non-ICU postoperative care, with almost no significant differences in morbidity, mortality or flap failure between the ICU and ward level postoperative care, but with lengthy hospital stay in those patients admitted to ICU [10][11].

Reviewing the literature, there is no clear consensus which post-operative free flap management protocol is better; an ICU admission versus non-ICU care. Due to lack of the evidence regarding which postoperative free flap nursing is associated with the highest flap survival and lower complications rates, therefore, the current systematic review and meta-analysis aimed to summarize the available evidence and to answer the question "Is there a difference in the flap failure and postoperative complication rates between the patients who postoperatively admitted to intensive care unit (ICU) or Non-ICU after microvascular head and neck reconstructive surgery?"

Methods

This systematic review and meta-analysis adhered to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement for reporting systematic reviews [12](Supplemental Table S1). The protocol of the meta-analysis has been registered on the PROSPERO platform (CRD42020207772).

Focused question

Is there a difference in the flap survival and postoperative complications rates between admission to intensive care unit (ICU) versus Non-ICU following microvascular head and neck reconstructive surgery?

The question for the current meta-analysis was adopted to follow PICO criteria:

P: Patients who underwent tumor resection and free flap reconstruction.

I: Participants who received nursing in ICU or non-ICU department.

C: ICU versus non-ICU admission

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O: flap survival rate reoperation, readmission, and respiratory complications, delirium, sepsis, Length of hospital stays, or Hospital cost.

T: nursing in ICU for ≥ 12 hour versus immediate nursing in non-ICU unit.

S: randomized controlled trail or observational studies.

Search strategies

From inception to April 2021, an electronic search without time or language restrictions was done in the following databases: PubMed, Embase, Scopus, and the Cochrane library (Supplemental Table S2). The following terms were used in the search strategy on PubMed:

(((((((head and neck neoplasms [MeSH Terms]) OR (microsurgical free flap [MeSH Terms])) AND (nursing, perioperative [MeSH Terms])) OR (specialized care unit)) AND (ward care)) OR (care, surgical intensive [MeSH Terms])) AND (postoperative complications [Title/Abstract])) OR (flap failure[Title/Abstract]).

A manual search of the following journals was also performed: International Journal of surgery, Journal of plastic and reconstructive surgery, Journal of plastic, esthetic and reconstructive surgery, Journal of annals of plastic surgery, Journal of Intensive care, Journal of Intensive care medicine, Journal of Intensive care, Journal of head and neck, Laryngoscope, British Journal of Oral and Maxillofacial Surgery, International Journal of Oral and Maxillofacial Surgery, Journal of Craniofacial Surgery, Journal of Cranio-Maxillofacial Surgery, Journal of Microsurgery, Oral Surgery Oral Medicine Oral Pathology Oral Radiology and Endodontology, Journal of head and neck surgery and JAMA otolaryngology head and neck surgery. The reference list of the identified studies and the relevant reviews on the subject were also evaluated for possible additional studies. Moreover, online databases providing information about clinical trials in progress were checked (clinicaltrials.gov; www. centerwatch.com/clinical trials; www.clinicalconnection.com). Details search strategy and number of yielded studies at different stages of screening are presented in the Figure 1 and supplementary file Table S2

Inclusion and Exclusion Criteria

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Eligibility criteria included human experimental or observational studies that compared the immediate postoperative admission of patients undergoing head and neck reconstructive surgery to ICU versus non-ICU and reported one of the following outcomes: Flap failure, reoperation, readmission, and respiratory complications, delirium, sepsis, Length of hospital stays, or Hospital cost. The exclusion criteria were other studies that reported one of the following: 1) Studies not comparing the postoperative ICU versus No-ICU care. 2) Studies with < 30 patients in each group. 3) Studies including an anticipated need for post-operative ICU care. 4) Studies including patients under the age of 18, 6) Review studies, meeting abstracts, and/or non-English articles.

Data extraction process

Two researchers independently assessed the titles, abstracts, and full-texts of the relevant studies and any controversy was resolved by discussion to reach a consensus. The following data were collected for each study (when available): authors, publication year, country of origin, study design, Age of the participants, number of patients in each group, Length of ICU admission, length of hospital stay (LOS), nursing ratios during study, total hospital charge, number of flap failure in each group, and other outcomes (Tables 1). Two researchers independently reviewed the included articles and collected the data. Disagreements between the reviewers were resolved by discussion between and if necessary, a third reviewer was involved to reach a consensus. A further contact with authors for possible missing data was also performed.

Patient and public involvement

The present study was a meta-analysis and systematic review based on published data. Patients and public were not involved in the study design, conduct, data analysis and result dissemination.

Methodological quality appraisal

Two authors independently assessed the risk of bias in the included studies. Quality assessment of the risk of bias for RCTs was carried out using Cochrane collaboration's tool [13], The RCTs were evaluated using the following six items: random sequence generation, allocation concealment, blinding of outcome assessment, incomplete outcome data, selected reporting, and other bias. Study met all the above criteria, the study was then rated as low risk of bias. If one or more of the above domains were unclear, the study

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was considered as unclear risk of bias. If one or more of these criteria were not met, the study was classified as having a high risk of bias. The ROBINS-I tools [14] were used for assessment of the non-RCTs. The non-RCT studies were assessed using the following seven domains: confounding factors, selection of participants, measurement of interventions, departures from intended interventions, missing data, bias in measurement of outcomes, and selection of reported results. The two independent authors then judged each study to be at low risk of bias, moderate risk of bias, serious risk of bias, critical risk of bias, or no information. (Table 2).

Statistical Analysis

For binary data, an Risk ratio (RR) was estimated. For continuous data, the mean difference (MD) was estimated. The weighted RR or MD along with 95% CI were used to construct forest plots of the selected studies. The I^2 statistic were used to express the percentage of the total variation across studies due to heterogeneity; I^2 with 25% corresponding to low heterogeneity, 50% to moderate, and 75% to high[15]. When I^2>50%, the random effect model was used as described by DerSimonian and Laird [16]. Otherwise, the data were regarded as homogeneous, and a fixed-effect model was used. A p-value of <0.05 was considered statistically significant[16]. Subgroup mete-analysis or meta-regression will be conducted to explore possible causes of heterogeneity among study results. If there were sufficient included studies, a sensitivity analysis will be performed to assess the robustness of our meta-analysis. This can be achieved by exclusion of studies with a high risk of bias. If there will be enough studies, funnel plot will be conducted to investigate publication and other bias in meta-analysis. A Comprehensive Meta-Analysis Software version 3 (Biostat Inc, Englewood, NJ) was utilized for data analysis.

Results

The kappa value was 0.85, so the agreement between investigators was almost perfect. The electronic and manual searches identified 2286 articles (Fig. 1); 1754 records remained after removal of the duplicates. The titles and abstracts of the remaining 1754 articles were screened and 1734 records were excluded due to being off topic or non-English studies. The remaining 20 studies were carefully read by two researchers for potential inclusion. Of the 20 full-text studies reviewed for potential inclusion; 8 met the inclusion

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criteria and were assessed for reliability (Table 1), with the other 12 articles excluded with reasons (Supplemental Table S3). The 8 included articles had a total of 2349 participants. One study was a prospective randomized controlled trials [17] and the other studies were nonrandomized studies[9,11,18–22]. Other general characteristics of the included studies were reported in the Table 1. The mean age of the patients ranged between 52- 64 years. The nursing ratios were reported in 4 studies [9,11,17–22] Table 1.

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Table 1 Characteristics of the included studies.

Study	Study design	Age (Years)	No. of Participants	Length of ICU admission	Nursing ratios	LOS (days)	Flap loss	Total hospital charges (mean)	outcomes
Nkenke et al. [18] 2009 Germany	PS	Mean (SD) ICU= 62.8 (9.7) Non-ICU= 63.5 (9.5)	ICU=50 Non-ICU=50	Mean (SD) 3.5 (10.9)	NR	Mean ICU=10.28 Non-ICU= 9.89	1	NR	Flap loss and other complications, Readmission, Length of hospital stays, and hospital cost
Arshad et al.**[9]** 2013 USA	RS	Mean ICU=59.2 Non-ICU=58.9	ICU=131 Non-ICU=126	NR	NR	Mean ICU=10.28 Non-ICU= 9.89	9	ICU was $2134 greater than non-ICU	Flap failure, LOS, hospital cost and other complications.
Yang et al. [20] 2019 USA	RS	Mean (SD) 64.28 (11.70)	ICU=205 Non-ICU=100	NR	NR	Mean (range) ICU= 6 (5–8). Non-ICU= 5 (5–7)	9	35% higher hospital costs in ICU	Flap loss, Reoperation, Readmission, LOS, other complications.
Panwar et al. **[11]** 2015 USA	RS	Mean (SD) ICU= 62.46(13.5) Non-ICU= 63.3 (11.6)	ICU=175 Non-ICU=72	NR	1:3 to 1:4	Mean (range) ICU=8 (7–11) Non-ICU= 7 (6–9.5)	9	Mean (range) ICU= $33,642 ($28,143–$43,196 ;Non-ICU= $28,524 ($22,611–	Flap failure, in-hospital mortality, and morbidity, hospital length of stay (LoS), total hospital-based charges, and cost of care, Operative time, ischemia time, site of tissue harvest.

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								$33,226)	
Chen et al. [21] 2018 Taiwan	RS	Mean (SD) ICU=53.8 ± 10.3 Non-ICU=52.2 ± 9.6	ICU=138 Non-ICU=179	NR	1:2 in the ICU; 1:9 during the day, 1:12 in the evening, and 1:16 in night shift in the ward	Mean (SD) ICU=22.1 ± 13.1 Non-ICU=24.2 ± 13.9	8	NR	ventilation length, use of sedation, flap outcomes, flap complications, systemic complications, in-hospital mortality, and hospitalization. Flap outcomes
Yalamanchi et al.[22] 2020 USA	RS	Mean (SD) ICU=64.6 (13.0) Non-ICU=61.5 (13.2)	ICU=146 Non-ICU= 192	Mean 7 (IQR 6-9) days	1:3	Mean (IQR) ICU=8.5 (6–9) Non-ICU=8 (4–8)	13	Mean ICU= $47,315.44 Non-ICU= $38,853.50	Postoperative complications, reoperation, and flap failure, ICU length of stay, total LOS, readmission rates, and hospital cost.
Cervenka et al. **[17]** 2019 USA	RCT	>18	ICU=57 Non-ICU= 61	ICU for 12–24 h	1:3 and 1:4	Mean (Range) ICU= 8.89 (4–30). Non-ICU= 8.97 (4–21).	1	Mean ICU= $466,199 Non-ICU= $486,150	LOS, flap failure rate, surgical and medical complications, and total cost of hospitalization
Clemens et al.[19] 2015, USA	RS	Mean(SD) 60 (14.1) years	680 ICU= 605 Non-ICU= 75	NR	NR	Non-ICU= 5.9 6(1.8) ICU= 9.56 (7.5)	10	NR	Flap failure rate, LOS, Minor and Major complications

LOS, length of hospital stay; PS, prospective study; RS, retrospective study; RCT, randomized controlled trail; ICU, intensive care unit; Non-ICU, Non-intensive care unit; NR, not reported; SD, Standard deviation

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Quality assessment

Based on the Cochrane collaboration's tool for risk of bias assessment, only one RCT [17] was included and was considered as having unclear risk of bias.

Based on the ROBINS-I tools for non-RCTs risk assessment, the overall risk of bias was moderate in five included studies [9,11,17–22], whereas two studies [18,19] were rated as having serious risk of bias.

At the pre-intervention stage, confounding bias was low in most of the studies, except for two studies where confounding variables were moderate Table 2. The lack of randomization is reflected in the moderate selection bias identified in the majority of the studies. For the post-intervention stage, bias due to missing information was not clearly stated in all studies. Measurement of outcomes was serious in two studies because the method of outcome assessment was not reported completely Table 2.

Results of individual outcome variables

Flap survival

Flap survival was reported in 8 studies [9,11,17–22] with a total of 2349 patients. A total of 60 flaps were lost. The flap survival rate was 97.19% in the ICU group and 97.89% in the non-ICU admission; however, a statistically insignificant difference was reported (Fixed, RR, 1.46; 95% CI, 0.80; 2.69, P = 0.231, I^2 = 0%) (Fig 2).

Reoperation and re-admission to the ICU

Reoperation was reported in 5 studies [11,17,20–22] with a total of 1253 patients. The incidence of reoperation was 12.34% in ICU and 9.44% in non-ICU admissions; however, a statistically insignificant difference was reported (Random, RR, 0.830; 95% CI, 0.540; 1.276, P = 0.397, I^2 = 53%) (Table 3). Readmission to ICU was reported in four studies [17,18,20,22] with a total of 861 patients. The incidence of readmission was 10.91% in ICU and 8.95% in non-ICU admissions; however, a statistically insignificant difference was reported (Fixed, RR, 0.922; 95% CI, 0.611; 1.392, P = 0.700, I^2 = 0%) (Table 3).

Medical complications

- The incidence of pneumonia was reported in four studies [17–20] (6.87% in ICU and 2.45% in non-ICU) with a total of 1203 patients. There was a statistically significant higher incidence of

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pneumonia in ICU compared to non-ICU nursing (Fixed, RR, 2.769; 95% CI, 1.193; 6.424, P = 0.018, I^2 = 0%) (Fig 3).

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Table 2. Results of the quality assessment for non-randomized studies using ROBINS-I tools

Study, Year	Pre-intervention	At intervention	Post-intervention					overall risk of bias
	Bias due to confounding	Bias in selection of participants into the study	Bias in classification of interventions	Bias due to deviations from intended interventions	Bias due to missing data	Bias in measurment of outcomes	Bias in selection of the reported result	
Nkenke et al. [18], 2009	Moderate	Low	Moderate	Moderate	Moderate	Serious	Moderate	Serious
Arshad et al.[9], 2013	Low	Low	Moderate	Moderate	Low	Moderate	Moderate	Moderate
Clemens et al.[19] 2015	Low	Low	Moderate	Moderate	Low	Serious	Moderate	Serious
Panwar et al. [11] 2015	Low	Low	Low	Low	Moderate	Moderate	Moderate	Moderate
Chen et al. [21] 2018	Moderate	Low	Moderate	Low	Moderate	Moderate	Low	Moderate
Yang et al. [20] 2019	Low	Low	Moderate	Moderate	Moderate	Moderate	Low	Moderate
Yalamanchi et al.[22] 2020	Low	Low	Moderate	Low	Low	Moderate	Moderate	Moderate

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Table 3. Summary of the results of the individual outcomes.

Outcome Variable	No. of study	No. of patients	Incidence rate (%)	Effect size Risk Ratio (RR); (95% CI)	Heterogeneity test	P value
Flap survival	8	2349	ICU=97.19 in Non-ICU=97.89%	RR, 1.46; CI, (0.80; 2.69)	0%	0.231
Partial flap necrosis	2	880	ICU=1.98 Non-ICU=2.4	RR, 1.251; CI,0.321; 4.876	0	0.747
Venous congestion	3	1097	ICU=2.90 Non-ICU=3.39	RR, 1.047; CI, 0.486; 2.257	0	0.906
Arterial thrombosis	3	1097	ICU=1.01 Non-ICU=1.97	RR, 0.804; CI,0.261; 2.471	0	0.703
Reoperation	5	1253	ICU= 12.34% non-ICU=9.44%	RR, 1.327; CI, 0.693; 2.539	53%	0.394
Readmission	4	861	ICU=10.91% Non-ICU=8.95%	0.922; 95% CI, 0.611; 1.392	0	0.700
Pneumonia	4	1203	ICU=6.87 Non-ICU=2.45	RR, 2.769; CI, 1.193;6.424	0	0.018
Respiratory failure	3	1027	ICU=11.33 Non-ICU=6.09	RR, 1.675; CI,0.911;3.078	35%	0.097
Need for ventilation	2	365	ICU=12.5 Non-ICU=10.52	RR, 1.926; CI, 1.128;3.288	0	0.016
Sepsis	5	1462	ICU=1.6 Non-ICU=0	RR, 4.23; CI, 1.12; 15.91	0	0.033
Delirium	2	365	ICU=12.5 Non-ICU=6.02	RR, 2.067; CI, 0.944; 44.526	0	0.069
Mortality	3	1115	ICU=0.38 Non-ICU=0.32	RR, 0.633; CI, 0.072; 5.532	0	0.679

ICU, Intensive Care Unit; Non-ICU, Non-Intensive Care Unit; RR, Risk Ratio; CI, confidence intervals.

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- The incidence of sepsis was reported in 5 studies [11,17–19,21] (1.6% in ICU and 0% in non-ICU) with a total of 1462 patients. There was a statistically significant higher incidence of sepsis in the ICU compared to non-ICU (Fixed, RR, 4.23; CI, 1.12; 15.91, P = 0.033, I2 = 0%) (Fig 4).
- No statistically significant difference was reported when comparing ICU and non-ICU concerning delirium, respiratory failure, and mortality (P> 0.05) (Table 3).

Length of hospital stay (LOS) and total hospital charge (THC)

LOS was reported in 4 studies [17,19,21,22] with a total of 773 patients. The reported LOS stay ranged from 4- 24 days Table 1. There was a statistically insignificant increase in LOS of about 1.5 day in the ICU group (Random, MD, 1.5; 95% CI, -0.835; 3.861, P = 0.270) (Fig 5). We were unable to conduct meta-analysis regarding hospital charge due to heterogenety and missing of some data. Cervenka et al[19], reported no significant difference between ICU and Non-ICU nursing regarding THC. Yang et al [20] stated that THC is about 35% higer in the patients admitted to ICU. Panwar et al[11] and Yalamanchi et al [22] reported a statistically significant difference between ICU and Non-ICU caring P<0.001 and 0.00054 respectively.

Discussion

The current systematic review and meta-analysis is novel in that it provides an evidence to suggest that patients who will undergo head and neck microvascular reconstructive surgery may not require routine postoperative ICU care, and the non-ICU nursing can provide equivalent clinical outcomes. This was consistent with several reported studies [9,11,18,23,24], however, this concept does not mean that those patients do not require a special care, but, it is feasible if an appropriate environment and adequately-trained nursing staff are available [25]. Bhama et al compared two types of ICU care (open versus closed) of patients with head and neck free flap surgeries and they did not report any significant differences in the postoperative outcomes between the two nursing policies. The current meta-analysis included 8 studies with a total of 2349 patients and showed a statistically insignificant difference between ICU and non-ICU

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admission concerning flap survival rate. The pooled flap survival rate was 97.19 % in the ICU group and 97.89% in the non-ICU group and this was consistent with the results obtained from other studies [24,26]. This result indicates that the postoperative admission in ICU or non-ICU settings does not influence flap survival rate. This was in line with the study performed by Chen et al, in that they showed no significant difference in the flap survival rate between ICU and non-ICU caring and they reported flap success rate of 96.7% and 98.3% respectively.

The common justification for routine postoperative care in the ICU of the patients who underwent microvascular free flap reconstruction is close flap monitoring and airway management [22]. However; close flap monitoring can also be achieved in the non- ICU setting; provided that this is done by an adequately trained nursing staff and in a controlled environment, and this includes an hourly physical examination of the flap to assess color, capillary refill, tissue temperature, turgor, pinprick test and Doppler signals. However, to improve the success rate of free flaps; particularly, in cases with poorly accessible flaps, an implantable Doppler would provide a higher flap salvage rate compared to the conventional monitoring method (81.4 % vs. 60.4%). [27].

On the other hand, to secure the airway after head and neck free flap reconstruction and simultaneously decreasing the need for postoperative ICU care, Nkenke et al[18] considered primary tracheostomy as a key point that allows avoiding the need for ICU admission. In a postal survey performed by Marsh et al.[7] in UK, they concluded that 69% of oral and maxillofacial surgeons prefer to perform tracheostomy after head and neck free flap surgery. However, tracheostomy is still not without complications (amounted of about 8-45%), [28,29] being chest infection the commonest complication (8%)[25].

The reported incidence of readmission to ICU in the present study for previously nursed patients in ICU or non-ICU setting was 10.91% and 8.95% respectively, and also no statistically significant difference was reported (P=0.7). These results indicated that patients underwent head and neck flap transfer might need readmission to ICU due to late postoperative complications, and these complications may happen regardless of the initial postoperative care in ICU or non-ICU setting.

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Some surgeons belief that one of the benefits of the ICU admission is to limit patient mobilization by using sedative medications and this would keep the vascular patency by preventing kinking of microvascular anastomoses and allows more invasive hemodynamic monitoring [9], however, an evidence supporting or conflicting this belief is still lacking. In addition, admission to ICU is not free of disadvantages including nosocomial infections, cancellation of operation due to limited bed availability, and increased cost [10]. Moreover, the use of sedation may decrease systemic blood pressure and subsequent reduction in blood perfusion of the flap [4].

Postoperative pulmonary complications are one of the most frequent complications in patients undergoing radical head and neck cancer surgery with free flap reconstruction, accounted for about 32% (pulmonary edema in 23.6% and pneumonia in 9.1%)[30]. Saied et al [31] conducted a large study on 14212 patients who were admitted to the ICU for more than 48 hour. They reported the incidence of two types of pneumonia in patients admitted in the ICU; the ventilator-associated pneumonia 1,161(15%) and ICU–hospital-acquired pneumonia 176 (2%). The current meta-analysis showed a statistically significant higher incidence of postoperative pneumonia and sepsis in the ICU nursing group compared to Non-ICU, P= 0.018 and 0.033 respectively. The pooled incidences of pneumonia in ICU and non-ICU were 6.4% and 2.8% respectively, meaning that patients admitted to ICU are at higher risk of development of ventilator-associated pneumonia or ICU–hospital-acquired pneumonia compared to non-ICU admitted patients.

We were unable to perform a meta-analysis concerning hospital charge because of the heterogeneity in the reported data. However, according to the existing literature, ICU care fee reported to be accounted for about 35% of the total hospital cost.[20] Therefore, admission to non-ICU care could reduce the high cost of ICU services and facilitates the management of available resources.

Although this systematic review and meta-analysis is the first to give a critical summary of the literature, pointing towards the unnecessity to nurse the carefully selected patients undergoing microvascular head and neck reconstructive surgery in the ICU setting. However, some limitations of this study should be declared. First, only one RCT was included; however, conducting a well-designed RCT in the field of reconstructive surgery appear to be difficult because there are many methodological and practical

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challenges of the surgical trials, such as recruitment, patient and surgeon preferences, feasibility of blinding, surgical quality control, and standardization of interventions. Second, several cofounders that might have an effect on the postoperative outcomes were not evaluated in the included studies such as the level and specialist training of the staff, nurse to patient's ratio, amount of intraoperative fluid used, use of vasopressors and anticoagulant … etc. For an optimum comparison of different nursing protocols with the least bias, homogeneous sample of patients with the same flap type and flap size, same recipient site, same perioperative care is needed. Third, although there was a consensus among the included studies in that ICU care increases the total hospital cost of care, no meta-analysis was performed due to both missing and heterogeneity in the reported data. Fourth, the non-English studies were excluded from the present study which may result in what is called language bias; however, several studies shown that exclusion of non-English publications from systematic reviews did not significantly affect the results of the meta-analyses [32,33]. In addition, studies published in non-English languages are difficult to locate, and may require translation, which will increase costs and delay the conclusion of a review.

Nevertheless, the current meta-analysis provides evidence regarding the feasibility of nursing the carefully selected patients undergoing head and neck free flap surgery in non-ICU setting.

Conclusion

The present systematic review and meta-analysis demonstrates that an immediate postoperative ICU admission of patients who have undergone microvascular head and neck reconstructive surgery did not reduce flap failure and postoperative complication rates. However, limiting the routine ICU admission to the carefully selected patients may result in a significant cost savings and reduction in the incidence of postoperative pneumonia and sepsis. Regardless where was the nursing of the patients who underwent head and neck microvascular flap surgery, the availability of an appropriate environment and adequately-trained nursing staff are the main key points and essential in the management of the postoperative complications.

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Footnotes

- **Contributors** MAM and TA conceived and designed the study. MAM and TA undertook data extraction and quality assessments. MAM, HA and ZSY performed statistical analyses. HA, TY and WLP prepared the figures and tables. BS, and NHA risk of bias assessment. All the authors provided inputs and were involved with the interpretation of the results. MAM and AA drafted the manuscript. MAM, AA, NHA, BS, TY, WLP and LHG revised the manuscript. All the authors have read and approved the final version.
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- **Data availability statement:** All data relevant to the study are included in the article or uploaded as supplementary information.

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Figures

Figure 1

Study flow diagram

Figure 2

Forest plot of flap survival rate. There was no difference in flap failure between ICU and non-ICU admission (Fixed, RR, 1.49; 95% CI, 0.80; 2.69, P = 0.182, I^2 = 0%). The horizontal lines indicate the 95% CI for the 8 included studies. The squares in the middle of the lines indicate the mean effect of the study. The diamond shape at the bottom indicates the 95% CI for the pooled effect.

Figure 3

Forest plot of the incidence of pneumonia. There was statistically significant higher incidence of pneumonia in ICU compared to non-ICU nursing (Fixed, RR, 2.769; 95% CI, 1.193; 6.424, P = 0.018, I^2 = 0%). The horizontal lines indicate the 95% CI for the 4 included studies. The squares in the middle of the lines indicate the mean effect of the study. The diamond shape at the bottom indicates the 95% CI for the pooled effect.

Figure 4

Forest plot of the incidence of sepsis. There was statistically significant higher incidence of sepsis in ICU compared to non-ICU nursing (Fixed, RR, 4.23; CI, 1.12; 15.91, P = 0.033, I2 = 0%). The horizontal lines indicate the 95% CI for the 5 included studies. The squares in the middle of the lines indicate the mean effect of the study. The diamond shape at the bottom indicates the 95% CI for the pooled effect.

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Figure 5

Forest plot of length of hospital stay (LOS). There was a statistically insignificant increase in LOS of about 1.5 day in the ICU group compared to non-ICU protocol. The horizontal lines indicate the 95% CI for the 4 included studies. The squares in the middle of the lines indicate the mean effect of the study. The diamond shape at the bottom indicates the 95% CI for the pooled effect.

Supplemental materials

Table S1. PRISMA checklist.

Table S2. Search terms and results of Search strategy

Table S3. Excluded study and reasons for exclusion

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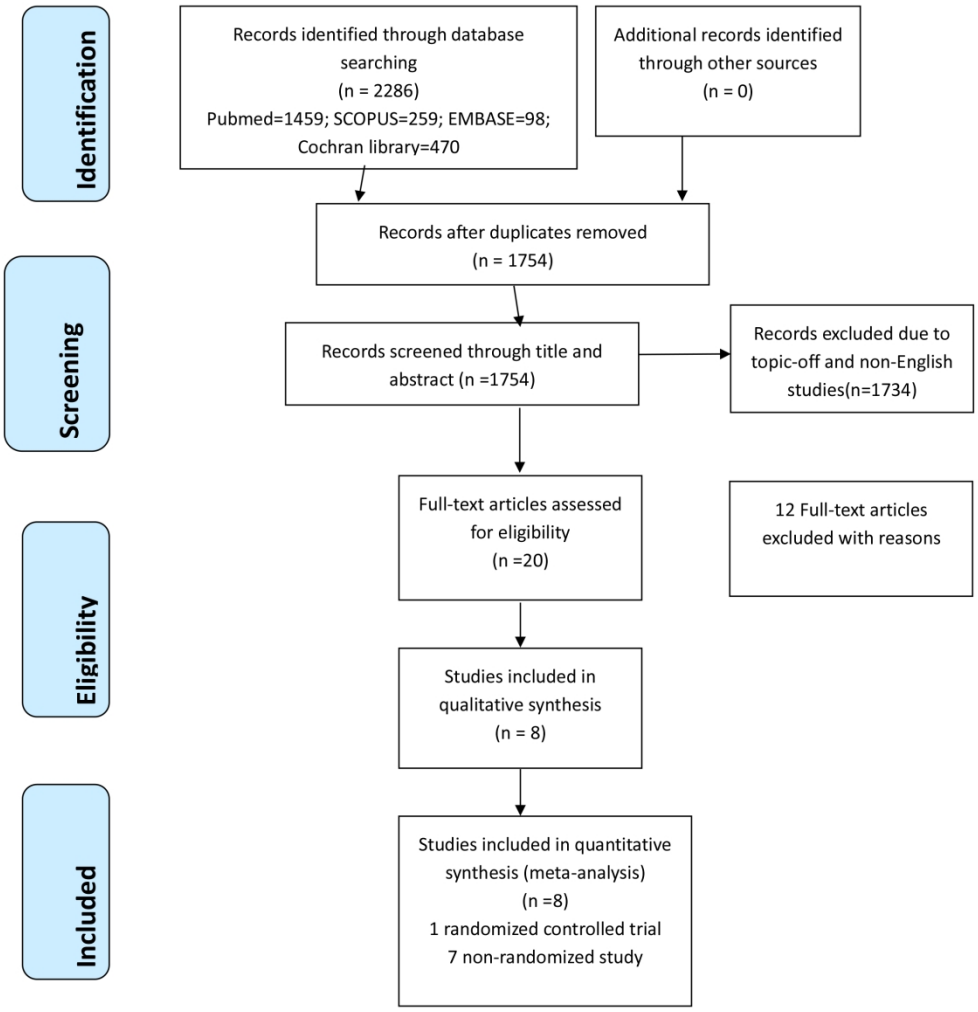


Figure 1. Study flow diagram

Figure 1
Study flow diagram

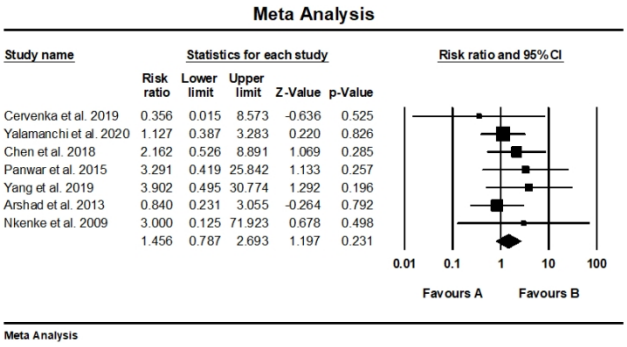


Figure 2
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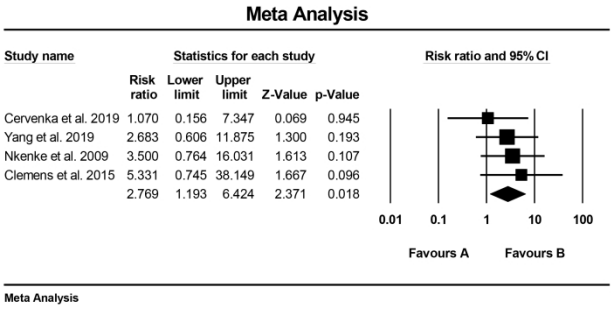


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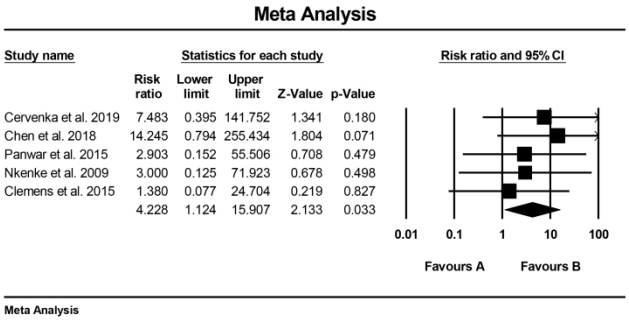


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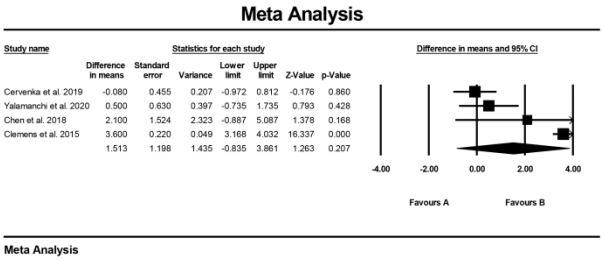


Figure 5
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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			1
Title	1	Identify the report as a systematic review.	1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	2
INTRODUCTION			3,4
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	3,4
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	4
METHODS			4-7
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	5,6
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	5
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	5
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	6
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	6
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	5,6,7
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	5
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	6,7
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	7
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	5,7
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	7
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	7
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	7
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	7
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	7
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	7
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	-

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	7,8
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	8
Study characteristics	17	Cite each included study and present its characteristics.	8, Table 1
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	10
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	11-15
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	11-15
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	11-15
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	-
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	-
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	-
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	-
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	15
	23b	Discuss any limitations of the evidence included in the review.	17,18
	23c	Discuss any limitations of the review processes used.	17,18
	23d	Discuss implications of the results for practice, policy, and future research.	18
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	4
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	4
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	4
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	19
Competing interests	26	Declare any competing interests of review authors.	19
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	-

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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Table S1: Search terms

	Search terms			No.
Pubmed 29-04-2021	(((((((head and neck neoplasms[MeSH Terms]) OR (microsurgical free flap[MeSH Terms])) AND (nursing, perioperative[MeSH Terms])) OR (specialized care unit)) AND (ward care)) OR (care, surgical intensive[MeSH Terms])) AND (postoperative complications[Title/Abstract])) OR (flap failure[Title/Abstract])			1459
SCOPUS 29-04-2021	head and neck tumor OR free flap reconstruction AND intensive care unit OR surgical intensive care unit AND intensive care unit OR surgical intensive care unit OR specialized care AND graft failure OR postoperative complications			259
EMBASE 29-04-2021	('head and neck tumor'/exp OR 'free flap reconstruction'/exp) AND ('intensive care unit'/exp OR 'surgical intensive care unit'/exp) AND ('intensive care unit'/exp OR 'surgical intensive care unit'/exp OR 'specialized care'/exp) AND ('graft failure'/exp OR 'postoperative complications'/exp)			98
Cochran Library 29-04-2021	#1	MeSH descriptor: [Head and Neck Neoplasms] explode all trees		470
	#2	specialized care unit		
	#3	MeSH descriptor: [Free Tissue Flaps] explode all trees		
	#4	MeSH descriptor: [Intensive Care Units] explode all trees		
	#5	MeSH descriptor: [Postoperative Complications] explode all trees		
		39311		
	#6	flap loss		
	#7	#1 OR #3 AND #2 AND #4 AND #6		
	#8	#7 AND #5		

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Table S2 Excluded studies and reason for exclusion

	Study	Reason for exclusion
1	Godden et al. 1999 [1]	Included patients who underwent tumor resection without flap reconstruction
2	Moreno and Bonilla-Velez[2]	Compared three different protocols. Two protocol admitted to the ICU for overnight before transfer to the intermediate care and to the general floor at physician discretion.
3	EDWARD et al. 2002 [3]	Although the author stated that some patients were unadmitted to the ICU, but no outcome were reported regarding patients in Non-ICU setting
4	Vimawala et al[4]	Conference poster
5	Morse et al 2019 [5]	All groups admitted to the ICU before transfer to the Non-ICU setting.
6	Yu et al. 2019 [6]	All patients were transferred to the intermediate level care.
7	Mathew et al 2010[7]	Only 7 patients were transferred to the ICU
8	Dautremont et al 2013[8]	Patients in both groups were admitted to the ICU.
9	Bhama et al 2013[9]	Patients were admitted to open or close ICU.
10	Patel et al. 2017 [10]	No clear information about the three-protocols used (ICU, intermediate level, and surgical ward).
11	Al-Dam et al. 2014[11]	No clear information about the two-protocol used (ICU, intermediate level). Flap failure or other outcomes of interest were not reported in both groups.
12	Bertelsen et al. 2020 [12]	Two different protocols were used but, both group were admitted to the ICU for overnight before transfer to the ward.

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Postoperative care in ICU versus non-ICU after head and neck free flap surgery: A systematic review and meta-analysis

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Postoperative care in ICU versus non-ICU after head and neck free flap surgery: A systematic review and meta-analysis

Short title: ICU versus non-ICU after free flap surgery

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Abstract

Objective Admission to the intensive care unit (ICU) has long been considered as routine by most head and neck surgeons after microvascular free flap transfer. This study aimed to answer the question "Is there a difference in the flap survival and postoperative complications rates between admission to intensive care unit (ICU) versus Non-ICU following microvascular head and neck reconstructive surgery?"**.**

Design Systematic review, and meta-analysis.

Methods the PubMed, Embase, Scopus, and Cochrane Library electronic databases were systematically searched (till April 2021) to identify the relevant studies. Studies that compared postoperative nursing of patients who underwent microvascular head and neck reconstructive surgery in ICU and non-ICU were included. The outcome variables were flap failure and length of hospital stay (LOS) and other complications. Weighted Odd ratio or mean differences with 95% confidence intervals were calculated.

Results Eight studies involving a total of 2349 patients were included. No statistically significant differences were observed between ICU and non-ICU admitted patients regarding flap survival reported (Fixed, RR, 1.46; 95% CI, 0.80; 2.69, P = 0.231, I^2 = 0%), reoperation, readmission, respiratory failure, delirium, and mortality (P>0.05). A significant increase in the postoperative pneumonia (P=0.018) and sepsis (P=0.033) was observed in patients admitted to ICU compared to non-ICU setting.

Conclusion This meta-analysis showed that an immediate postoperative nursing in the ICU after head and neck microvascular reconstructive surgery did not reduce the incidence of flap failure or complications rate. Limiting the routine ICU admission to the carefully selected patients may result in a reduction in the incidence of postoperative pneumonia, sepsis, LOS and total hospital charge.

Keywords**:** *Microvascular free flap; intensive care unit; flap survival; complications; postoperative care.*

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Strengths and limitations of this study

- *This is the first systematic review and meta-analysis to evaluate the postoperative outcomes of nursing in ICU versus non- ICU after head and neck microvascular reconstructive surgery.*
- *We analyzed data from both randomized trials and observational studies.*
- *For some outcomes, the numbers of included studies were few.*
- *The major limitation of this study was the high heterogeneity Between the published studies.*

Introduction

Recently, microsurgical procedures including Free flap reconstruction have gained a wide popularity becoming the standard of care for various surgical defects of the head and neck region [1]. Free flap surgery with various available flaps having different characteristics covers a wide range of indications in reconstructive head and neck surgery with different perspectives and an overall success rate reported up to 95% [2].

Being considered as a major surgery with long intraoperative time, the postoperative care of free flap surgery is considered critical and requires great attention. In addition, the need to closely monitor the flap viability for any vascular compromise occurring mostly in the first 48 to 72 hours, increases the need for an accurate and close monitoring in the immediate post-operative interval, therefore this interval is critical and of utmost importance for successful salvage of ischemic flaps [3]. Nevertheless, still which postoperative protocol is the most suitable is an issue of controversy and there is no international consensus between surgeons regarding a common monitoring regimen [4].

For years, admission to the intensive care unit (ICU) has long been considered as routine by most head and neck reconstructive surgeons following free flap surgery with a mean of 2.4 days admission in the ICU [5]. It was reported that 88.9% of surgeons prefer sending their free flap patients to ICU even though most patients does not meet the cardiopulmonary indications of ICU admission [6][7]. In historic literature,

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certain post-operative treatment regimens stated complete patient immobilization during the immediate post-operative period in an effort to protect the vascular pedicle [8]. This has evolved to the most common accepted standard where patients are spontaneously breathing at the conclusion of surgery. However, admission to the ICU is not without drawbacks due to prolonged mechanical ventilation, late ambulation, overuse of sedation, in addition to the higher incidence of pneumonia [8][9]. Accordingly, a different era of postoperative care of free flap patients has been evolved recently in practice entailing the postoperative care at ward level or non-ICU postoperative care, with almost no significant differences in morbidity, mortality or flap failure between the ICU and ward level postoperative care, but with lengthy hospital stay in those patients admitted to ICU [10][11]. Reviewing the literature, there is no clear consensus which post-operative free flap management protocol is better; an ICU admission versus non-ICU care. Due to lack of the evidence regarding which postoperative free flap nursing is associated with the highest flap survival and lower complications rates, therefore, the current systematic review and meta-analysis aimed to summarize the available evidence and to answer the question "Is there a difference in the flap failure and postoperative complication rates between the patients who postoperatively admitted to intensive care unit (ICU) or Non-ICU after microvascular head and neck reconstructive surgery?"

Methods

This systematic review and meta-analysis adhered to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement for reporting systematic reviews [12](Supplemental Table S1). The protocol of the meta-analysis has been registered on the PROSPERO platform (CRD42020207772).

Focused question

Is there a difference in the flap survival and postoperative complications rates between admission to intensive care unit (ICU) versus Non-ICU following microvascular head and neck reconstructive surgery?

The question for the current meta-analysis was adopted to follow PICOTS criteria:

P: Patients who underwent tumor resection and free flap reconstruction.

I: Participants who received nursing in ICU or non-ICU department.

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C: ICU versus non-ICU admission

O: flap survival rate reoperation, readmission, and respiratory complications, delirium, sepsis, Length of hospital stays, or Hospital cost.

T: nursing in ICU for ≥ 12 hours versus immediate nursing in non-ICU unit.

S: randomized controlled trial or observational studies.

Search strategies

From inception to April 2021, an electronic search without time or language restrictions was done in the following databases: PubMed, Embase, Scopus, and the Cochrane library (Supplemental Table S2). The following terms were used in the search strategy on PubMed:

(((((((head and neck neoplasms [MeSH Terms]) OR (microsurgical free flap [MeSH Terms])) AND (nursing, perioperative [MeSH Terms])) OR (specialized care unit)) AND (ward care)) OR (care, surgical intensive [MeSH Terms])) AND (postoperative complications [Title/Abstract])) OR (flap failure[Title/Abstract]).

A manual search of the following journals was also performed: International Journal of surgery, Journal of plastic and reconstructive surgery, Journal of plastic, esthetic and reconstructive surgery, Journal of annals of plastic surgery, Journal of Intensive care, Journal of Intensive care medicine, Journal of Intensive care, Journal of head and neck, Laryngoscope, British Journal of Oral and Maxillofacial Surgery, International Journal of Oral and Maxillofacial Surgery, Journal of Craniofacial Surgery, Journal of Cranio-Maxillofacial Surgery, Journal of Microsurgery, Oral Surgery Oral Medicine Oral Pathology Oral Radiology and Endodontology, Journal of head and neck surgery and JAMA otolaryngology head and neck surgery. The reference list of the identified studies and the relevant reviews on the subject were also evaluated for possible additional studies. Moreover, online databases providing information about clinical trials in progress were checked (clinicaltrials.gov; www. centerwatch.com/clinical trials; www.clinicalconnection.com). Details search strategy and number of yielded studies at different stages of screening are presented in the Figure 1 and supplementary file Table S2

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Inclusion and Exclusion Criteria

Eligibility criteria included human experimental or observational studies that compared the immediate postoperative admission of patients undergoing head and neck reconstructive surgery to ICU versus non-ICU and reported one of the following outcomes: Flap failure, reoperation, readmission, and respiratory complications, delirium, sepsis, Length of hospital stays, or Hospital cost. The exclusion criteria were other studies that reported one of the following: 1) Studies not comparing the postoperative ICU versus No-ICU care. 2) Studies with < 30 patients in each group. 3) Studies including an anticipated need for post-operative ICU care. 4) Studies including patients under the age of 18, 6) Review studies, meeting abstracts, and/or non-English articles.

Data extraction process

Two researchers independently assessed the titles, abstracts, and full-texts of the relevant studies and any controversy was resolved by discussion to reach a consensus. The following data were collected for each study (when available): authors, publication year, country of origin, study design, Age of the participants, number of patients in each group, Length of ICU admission, length of hospital stay (LOS), nursing ratios during study, total hospital charge, number of flap failure in each group, and other outcomes (Tables 1). Two researchers independently reviewed the included articles and collected the data. Disagreements between the reviewers were resolved by discussion between and if necessary, a third reviewer was involved to reach a consensus. A further contact with authors for possible missing data was also performed.

Patient and public involvement

The present study was a meta-analysis and systematic review based on published data. Patients and public were not involved in the study design, conduct, data analysis and result dissemination.

Methodological quality appraisal

Two authors independently assessed the risk of bias in the included studies. Quality assessment of the risk of bias for RCTs was carried out using Cochrane collaboration's tool [13], The RCTs were evaluated using the following six items: random sequence generation, allocation concealment, blinding of outcome assessment, incomplete outcome data, selected reporting, and other bias. Study met all the above criteria;

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the study was then rated as low risk of bias. If one or more of the above domains were unclear, the study was considered as unclear risk of bias. If one or more of these criteria were not met, the study was classified as having a high risk of bias. The ROBINS-I tools [14] were used for assessment of the non-RCTs. The non-RCT studies were assessed using the following seven domains: confounding factors, selection of participants, measurement of interventions, departures from intended interventions, missing data, bias in measurement of outcomes, and selection of reported results. The two independent authors then judged each study to be at low risk of bias, moderate risk of bias, serious risk of bias, critical risk of bias, or no information. (Table 2).

Statistical Analysis

For binary data, a Risk ratio (RR) was estimated. For continuous data, the mean difference (MD) was estimated. The weighted RR or MD along with 95% CI were used to construct forest plots of the selected studies. The I^2 statistic were used to express the percentage of the total variation across studies due to heterogeneity; I^2 with 25% corresponding to low heterogeneity, 50% to moderate, and 75% to high[15]. When I^2>50%, the random effect model was used as described by DerSimonian and Laird [16]. Otherwise, the data were regarded as homogeneous, and a fixed-effect model was used. A p-value of <0.05 was considered statistically significant[16]. Subgroup mete-analysis or meta-regression will be conducted to explore possible causes of heterogeneity among study results. If there were sufficient included studies, a sensitivity analysis will be performed to assess the robustness of our meta-analysis. This can be achieved by exclusion of studies with a high risk of bias. If there will be enough studies, funnel plot will be conducted to investigate publication and other bias in meta-analysis. A Comprehensive Meta-Analysis Software version 3 (Biostat Inc, Englewood, NJ) was utilized for data analysis.

Results

The kappa value was 0.85, so the agreement between investigators was almost perfect. The electronic and manual searches identified 2286 articles (Fig. 1); 1754 records remained after removal of the duplicates. The titles and abstracts of the remaining 1754 articles were screened and 1734 records were excluded due to being off topic or non-English studies. The remaining 20 studies were carefully read by two researchers

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for potential inclusion. Of the 20 full-text studies reviewed for potential inclusion; 8 met the inclusion criteria and were assessed for reliability (Table 1), with the other 12 articles excluded with reasons (Supplemental Table S3). The 8 included articles had a total of 2349 participants. One study was a prospective randomized controlled trials [17] and the other studies were nonrandomized studies[9,11,18–22]. Other general characteristics of the included studies were reported in the Table 1**.** The mean age of the patients ranged between 52- 64 years. The nursing ratios were reported in 4 studies [9,11,17–22] Table 1.

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Table 1 Characteristics of the included studies.

Study	Study design	Age (Years)	No. of Participants	Length of ICU admission	Nursing ratios	LOS (days)	Flap loss	Total hospital charges (mean)	outcomes
Nkenke et al. [18] 2009 Germany	PS	Mean (SD) ICU= 62.8 (9.7) Non-ICU= 63.5 (9.5)	ICU=50 Non-ICU=50	Mean (SD) 3.5 (10.9)	NR	Mean ICU=10.28 Non-ICU= 9.89	1	NR	Flap loss and other complications, Readmission, Length of hospital stays, and hospital cost
Arshad et al.**[9]** 2013 USA	RS	Mean ICU=59.2 Non-ICU=58.9	ICU=131 Non-ICU=126	NR	NR	Mean ICU=10.28 Non-ICU= 9.89	9	ICU was $2134 greater than non-ICU	Flap failure, LOS, hospital cost and other complications.
Yang et al. [20] 2019 USA	RS	Mean (SD) 64.28 (11.70)	ICU=205 Non-ICU=100	NR	NR	Mean (range) ICU= 6 (5–8). Non-ICU= 5 (5–7)	9	35% higher hospital costs in ICU	Flap loss, Reoperation, Readmission, LOS, other complications.
Panwar et al. **[11]** 2015 USA	RS	Mean (SD) ICU= 62.46(13.5) Non-ICU= 63.3 (11.6)	ICU=175 Non-ICU=72	NR	1:3 to 1:4	Mean (range) ICU=8 (7–11) Non-ICU= 7 (6–9.5)	9	Mean (range) ICU= $33,642 ($28,143–$43,196 ;Non-ICU= $28,524 ($22,611–	Flap failure, in-hospital mortality, and morbidity, hospital length of stay (LoS), total hospital-based charges, and cost of care, Operative time, ischemia time, site of tissue harvest.

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								$33,226)	
Chen et al. [21] 2018 Taiwan	RS	Mean (SD) ICU=53.8 ± 10.3 Non-ICU=52.2 ± 9.6	ICU=138 Non-ICU=179	NR	1:2 in the ICU; 1:9 during the day, 1:12 in the evening, and 1:16 in night shift in the ward	Mean (SD) ICU=22.1 ± 13.1 Non-ICU=24.2 ± 13.9	8	NR	ventilation length, use of sedation, flap outcomes, flap complications, systemic complications, in-hospital mortality, and hospitalization. Flap outcomes
Yalamanchi et al.[22] 2020 USA	RS	Mean (SD) ICU=64.6 (13.0) Non-ICU=61.5 (13.2)	ICU=146 Non-ICU= 192	Mean 7 (IQR 6-9) days	1:3	Mean (IQR) ICU=8.5 (6–9) Non-ICU=8 (4–8)	13	Mean ICU= $47,315.44 Non-ICU= $38,853.50	Postoperative complications, reoperation, and flap failure, ICU length of stay, total LOS, readmission rates, and hospital cost.
Cervenka et al. **[17]** 2019 USA	RCT	>18	ICU=57 Non-ICU= 61	ICU for 12–24 h	1:3 and 1:4	Mean (Range) ICU= 8.89 (4–30). Non-ICU= 8.97 (4–21).	1	Mean ICU= $466,199 Non-ICU= $486,150	LOS, flap failure rate, surgical and medical complications, and total cost of hospitalization
Clemens et al.[19] 2015, USA	RS	Mean(SD) 60 (14.1) years	680 ICU= 605 Non-ICU= 75	NR	NR	Non-ICU= 5.9 6(1.8) ICU= 9.56 (7.5)	10	NR	Flap failure rate, LOS, Minor and Major complications

LOS, length of hospital stay; PS, prospective study; RS, retrospective study; RCT, randomized controlled trail; ICU, intensive care unit; Non-ICU, Non-intensive care unit; NR, not reported; SD, Standard deviation

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Quality assessment

Based on the Cochrane collaboration's tool for risk of bias assessment, only one RCT [17] was included and was considered as having unclear risk of bias.

Based on the ROBINS-I tools for non-RCTs risk assessment, the overall risk of bias was moderate in five included studies [9,11,17–22], whereas two studies [18,19] were rated as having serious risk of bias.

At the pre-intervention stage, confounding bias was low in most of the studies, except for two studies where confounding variables were moderate Table 2. The lack of randomization is reflected in the moderate selection bias identified in the majority of the studies. For the post-intervention stage, bias due to missing information was not clearly stated in all studies. Measurement of outcomes was serious in two studies because the method of outcome assessment was not reported completely Table 2.

Results of individual outcome variables

Flap survival

Flap survival was reported in 8 studies [9,11,17–22] with a total of 2349 patients. A total of 60 flaps were lost. The flap survival rate was 97.19% in the ICU group and 97.89% in the non-ICU admission; however, a statistically insignificant difference was reported (Fixed, RR, 1.46; 95% CI, 0.80; 2.69, P = 0.231, I^2 = 0%) (Fig 2).

Reoperation and re-admission to the ICU

Reoperation was reported in 5 studies [11,17,20–22] with a total of 1253 patients. The incidence of reoperation was 12.34% in ICU and 9.44% in non-ICU admissions; however, a statistically insignificant difference was reported (Random, RR, 0.830; 95% CI, 0.540; 1.276, P = 0.397, I^2 = 53%) (Table 3). Readmission to ICU was reported in four studies [17,18,20,22] with a total of 861 patients. The incidence of readmission was 10.91% in ICU and 8.95% in non-ICU admissions; however, a statistically insignificant difference was reported (Fixed, RR, 0.922; 95% CI, 0.611; 1.392, P = 0.700, I^2 = 0%) (Table 3).

Medical complications

- The incidence of pneumonia was reported in four studies [17–20] (6.87% in ICU and 2.45% in non-ICU) with a total of 1203 patients. There was a statistically significant higher incidence of

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pneumonia in ICU compared to non-ICU nursing (Fixed, RR, 2.769; 95% CI, 1.193; 6.424, P = 0.018, I^2 = 0%) (Fig 3).

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Table 2. Results of the quality assessment for non-randomized studies using ROBINS-I tools

Study, Year	Pre-intervention	At intervention	Post-intervention					overall risk of bias
	Bias due to confounding	Bias in selection of participants into the study	Bias in classification of interventions	Bias due to deviations from intended interventions	Bias due to missing data	Bias in measurment of outcomes	Bias in selection of the reported result	
Nkenke et al. [18], 2009	Moderate	Low	Moderate	Moderate	Moderate	Serious	Moderate	Serious
Arshad et al.[9], 2013	Low	Low	Moderate	Moderate	Low	Moderate	Moderate	Moderate
Clemens et al.[19] 2015	Low	Low	Moderate	Moderate	Low	Serious	Moderate	Serious
Panwar et al. [11] 2015	Low	Low	Low	Low	Moderate	Moderate	Moderate	Moderate
Chen et al. [21] 2018	Moderate	Low	Moderate	Low	Moderate	Moderate	Low	Moderate
Yang et al. [20] 2019	Low	Low	Moderate	Moderate	Moderate	Moderate	Low	Moderate
Yalamanchi et al.[22] 2020	Low	Low	Moderate	Low	Low	Moderate	Moderate	Moderate

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Table 3. Summary of the results of the individual outcomes.

Outcome Variable	No. of study	No. of patients	Incidence rate (%)	Effect size Risk Ratio (RR); (95% CI)	Heterogeneity test	P value
Flap survival	8	2349	ICU=97.19 in Non-ICU=97.89%	RR, 1.46; CI, (0.80; 2.69)	0%	0.231
Partial flap necrosis	2	880	ICU=1.98 Non-ICU=2.4	RR, 1.251; CI,0.321; 4.876	0	0.747
Venous congestion	3	1097	ICU=2.90 Non-ICU=3.39	RR, 1.047; CI, 0.486; 2.257	0	0.906
Arterial thrombosis	3	1097	ICU=1.01 Non-ICU=1.97	RR, 0.804; CI,0.261; 2.471	0	0.703
Reoperation	5	1253	ICU= 12.34% non-ICU=9.44%	RR, 1.327; CI, 0.693; 2.539	53%	0.394
Readmission	4	861	ICU=10.91% Non-ICU=8.95%	0.922; 95% CI, 0.611; 1.392	0	0.700
Pneumonia	4	1203	ICU=6.87 Non-ICU=2.45	RR, 2.769; CI, 1.193;6.424	0	0.018
Respiratory failure	3	1027	ICU=11.33 Non-ICU=6.09	RR, 1.675; CI,0.911;3.078	35%	0.097
Need for ventilation	2	365	ICU=12.5 Non-ICU=10.52	RR, 1.926; CI, 1.128;3.288	0	0.016
Sepsis	5	1462	ICU=1.6 Non-ICU=0	RR, 4.23; CI, 1.12; 15.91	0	0.033
Delirium	2	365	ICU=12.5 Non-ICU=6.02	RR, 2.067; CI, 0.944; 44.526	0	0.069
Mortality	3	1115	ICU=0.38 Non-ICU=0.32	RR, 0.633; CI, 0.072; 5.532	0	0.679

ICU, Intensive Care Unit; Non-ICU, Non-Intensive Care Unit; RR, Risk Ratio; CI, confidence intervals.

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- The incidence of sepsis was reported in 5 studies [11,17–19,21] (1.6% in ICU and 0% in non-ICU) with a total of 1462 patients. There was a statistically significant higher incidence of sepsis in the ICU compared to non-ICU (Fixed, RR, 4.23; CI, 1.12; 15.91, P = 0.033, I2 = 0%) (Fig 4).
- No statistically significant difference was reported when comparing ICU and non-ICU concerning delirium, respiratory failure, and mortality (P> 0.05) (Table 3).

Length of hospital stay (LOS) and total hospital charge (THC)

LOS was reported in 4 studies [17,19,21,22] with a total of 773 patients. The reported LOS stay ranged from 4- 24 days Table 1. There was a statistically insignificant increase in LOS of about 1.5 day in the ICU group (Random, MD, 1.5; 95% CI, -0.835; 3.861, P = 0.270) (Fig 5). We were unable to conduct meta-analysis regarding hospital charge due to heterogenety and missing of some data. Cervenka et al[19], reported no significant difference between ICU and Non-ICU nursing regarding THC. Yang et al [20] stated that THC is about 35% higer in the patients admitted to ICU. Panwar et al[11] and Yalamanchi et al [22] reported a statistically significant difference between ICU and Non-ICU caring P<0.001 and 0.00054 respectively.

Discussion

The current systematic review and meta-analysis is novel in that it provides evidence to suggest that patients who will undergo head and neck microvascular reconstructive surgery may not require routine postoperative ICU care, and the non-ICU nursing can provide equivalent clinical outcomes. This was consistent with several reported studies [9,11,18,23,24], however, this concept does not mean that those patients do not require a special care, but, it is feasible if an appropriate environment and adequately-trained nursing staff are available [25]. Bhama et al compared two types of ICU care (open versus closed) of patients with head and neck free flap surgeries and they did not report any significant differences in the postoperative outcomes between the two nursing policies. The current meta-analysis included 8 studies with a total of 2349 patients and showed a statistically insignificant difference between ICU and non-ICU

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admission concerning flap survival rate. The pooled flap survival rate was 97.19 % in the ICU group and 97.89% in the non-ICU group and this was consistent with the results obtained from other studies [24,26]. This result indicates that the postoperative admission in ICU or non-ICU settings does not influence flap survival rate. This was in line with the study performed by Chen et al, in that they showed no significant difference in the flap survival rate between ICU and non-ICU caring and they reported flap success rate of 96.7% and 98.3% respectively.

The common justification for routine postoperative care in the ICU of the patients who underwent microvascular free flap reconstruction is close flap monitoring and airway management [22]. However; close flap monitoring can also be achieved in the non- ICU setting; provided that this is done by an adequately trained nursing staff and in a controlled environment, and this includes an hourly physical examination of the flap to assess color, capillary refill, tissue temperature, turgor, pinprick test and Doppler signals. However, to improve the success rate of free flaps; particularly, in cases with poorly accessible flaps, an implantable Doppler would provide a higher flap salvage rate compared to the conventional monitoring method (81.4 % vs. 60.4%). [27].

On the other hand, to secure the airway after head and neck free flap reconstruction and simultaneously decreasing the need for postoperative ICU care, Nkenke et al[18] considered primary tracheostomy as a key point that allows avoiding the need for ICU admission. In a postal survey performed by Marsh et al.[7] in UK, they concluded that 69% of oral and maxillofacial surgeons prefer to perform tracheostomy after head and neck free flap surgery. However, tracheostomy is still not without complications (amounted of about 8-45%), [28,29] being chest infection the commonest complication (8%)[25].

The reported incidence of readmission to ICU in the present study for previously nursed patients in ICU or non-ICU setting was 10.91% and 8.95% respectively, and also no statistically significant difference was reported (P=0.7). These results indicated that patients underwent head and neck flap transfer might need readmission to ICU due to late postoperative complications, and these complications may happen regardless of the initial postoperative care in ICU or non-ICU setting.

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Some surgeons belief that one of the benefits of the ICU admission is to limit patient mobilization by using sedative medications and this would keep the vascular patency by preventing kinking of microvascular anastomoses and allows more invasive hemodynamic monitoring [9], however, an evidence supporting or conflicting this belief is still lacking. In addition, admission to ICU is not free of disadvantages including nosocomial infections, cancellation of operation due to limited bed availability, and increased cost [10]. Moreover, the use of sedation may decrease systemic blood pressure and subsequent reduction in blood perfusion of the flap [4].

Postoperative pulmonary complications are one of the most frequent complications in patients undergoing radical head and neck cancer surgery with free flap reconstruction, accounted for about 32% (pulmonary edema in 23.6% and pneumonia in 9.1%)[30]. Saied et al [31] conducted a large study on 14212 patients who were admitted to the ICU for more than 48 hour. They reported the incidence of two types of pneumonia in patients admitted in the ICU; the ventilator-associated pneumonia 1,161(15%) and ICU–hospital-acquired pneumonia 176 (2%). The current meta-analysis showed a statistically significant higher incidence of postoperative pneumonia and sepsis in the ICU nursing group compared to Non-ICU, P= 0.018 and 0.033 respectively. The pooled incidences of pneumonia in ICU and non-ICU were 6.4% and 2.8% respectively, meaning that patients admitted to ICU are at higher risk of development of ventilator-associated pneumonia or ICU–hospital-acquired pneumonia compared to non-ICU admitted patients.

We were unable to perform a meta-analysis concerning hospital charge because of the heterogeneity in the reported data. However, according to the existing literature, ICU care fee reported to be accounted for about 35% of the total hospital cost.[20] Therefore, admission to non-ICU care could reduce the high cost of ICU services and facilitates the management of available resources.

Although this systematic review and meta-analysis is the first to give a critical summary of the literature, pointing towards the unnecessity to nurse the carefully selected patients undergoing microvascular head and neck reconstructive surgery in the ICU setting. However, some limitations of this study should be declared. First, only one RCT was included; however, conducting a well-designed RCT in the field of reconstructive surgery appear to be difficult because there are many methodological and practical

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challenges of the surgical trials, such as recruitment, patient and surgeon preferences, feasibility of blinding, surgical quality control, and standardization of interventions. Second, several cofounders that might have an effect on the postoperative outcomes were not evaluated in the included studies such as the level and specialist training of the staff, nurse to patient's ratio, amount of intraoperative fluid used, use of vasopressors and anticoagulant … etc. For an optimum comparison of different nursing protocols with the least bias, homogeneous sample of patients with the same flap type and flap size, same recipient site, same perioperative care is needed. Third, although there was a consensus among the included studies in that ICU care increases the total hospital cost of care, no meta-analysis was performed due to both missing and heterogeneity in the reported data. Fourth, the non-English studies were excluded from the present study which may result in what is called language bias; however, several studies shown that exclusion of non-English publications from systematic reviews did not significantly affect the results of the meta-analyses [32,33]. In addition, studies published in non-English languages are difficult to locate, and may require translation, which will increase costs and delay the conclusion of a review.

Nevertheless, the current meta-analysis provides evidence regarding the feasibility of nursing the carefully selected patients undergoing head and neck free flap surgery in non-ICU setting.

Conclusion

The present systematic review and meta-analysis demonstrates that an immediate postoperative ICU admission of patients who have undergone microvascular head and neck reconstructive surgery did not reduce flap failure and postoperative complication rates. However, limiting the routine ICU admission to the carefully selected patients may result in a significant cost savings and reduction in the incidence of postoperative pneumonia and sepsis. Regardless where was the nursing of the patients who underwent head and neck microvascular flap surgery, the availability of an appropriate environment and adequately-trained nursing staff are the main key points and essential in the management of the postoperative complications.

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Footnotes

- **Contributors** MAM and TA conceived and designed the study. MAM and TA undertook data extraction and quality assessments. MAM, HA and ZSY performed statistical analyses. HA, TY and WLP prepared the figures and tables. BS, and NHA risk of bias assessment. All the authors provided inputs and were involved with the interpretation of the results. MAM and AA drafted the manuscript. MAM, AA, NHA, BS, TY, WLP and LHG revised the manuscript. All the authors have read and approved the final version.
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- **Ethics approval:** Not applicable. This article does not contain any studies with human participants performed by any of the authors.
- **Competing interests** None declared.
- **Data availability statement:** All data relevant to the study are included in the article or uploaded as supplementary information.

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Figures

Figure 1

Study flow diagram

Figure 2

Forest plot of flap survival rate. There was no difference in flap failure between ICU and non-ICU admission (Fixed, RR, 1.49; 95% CI, 0.80; 2.69, P = 0.182, I^2 = 0%). The horizontal lines indicate the 95% CI for the 8 included studies. The squares in the middle of the lines indicate the mean effect of the study. The diamond shape at the bottom indicates the 95% CI for the pooled effect.

Figure 3

Forest plot of the incidence of pneumonia. There was statistically significant higher incidence of pneumonia in ICU compared to non-ICU nursing (Fixed, RR, 2.769; 95% CI, 1.193; 6.424, P = 0.018, I^2 = 0%). The horizontal lines indicate the 95% CI for the 4 included studies. The squares in the middle of the lines indicate the mean effect of the study. The diamond shape at the bottom indicates the 95% CI for the pooled effect.

Figure 4

Forest plot of the incidence of sepsis. There was statistically significant higher incidence of sepsis in ICU compared to non-ICU nursing (Fixed, RR, 4.23; CI, 1.12; 15.91, P = 0.033, I2 = 0%). The horizontal lines indicate the 95% CI for the 5 included studies. The squares in the middle of the lines indicate the mean effect of the study. The diamond shape at the bottom indicates the 95% CI for the pooled effect.

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Figure 5

Forest plot of length of hospital stay (LOS). There was a statistically insignificant increase in LOS of about 1.5 day in the ICU group compared to non-ICU protocol. The horizontal lines indicate the 95% CI for the 4 included studies. The squares in the middle of the lines indicate the mean effect of the study. The diamond shape at the bottom indicates the 95% CI for the pooled effect.

Supplemental materials

Table S1. PRISMA checklist.

Table S2. Search terms and results of Search strategy

Table S3. Excluded study and reasons for exclusion

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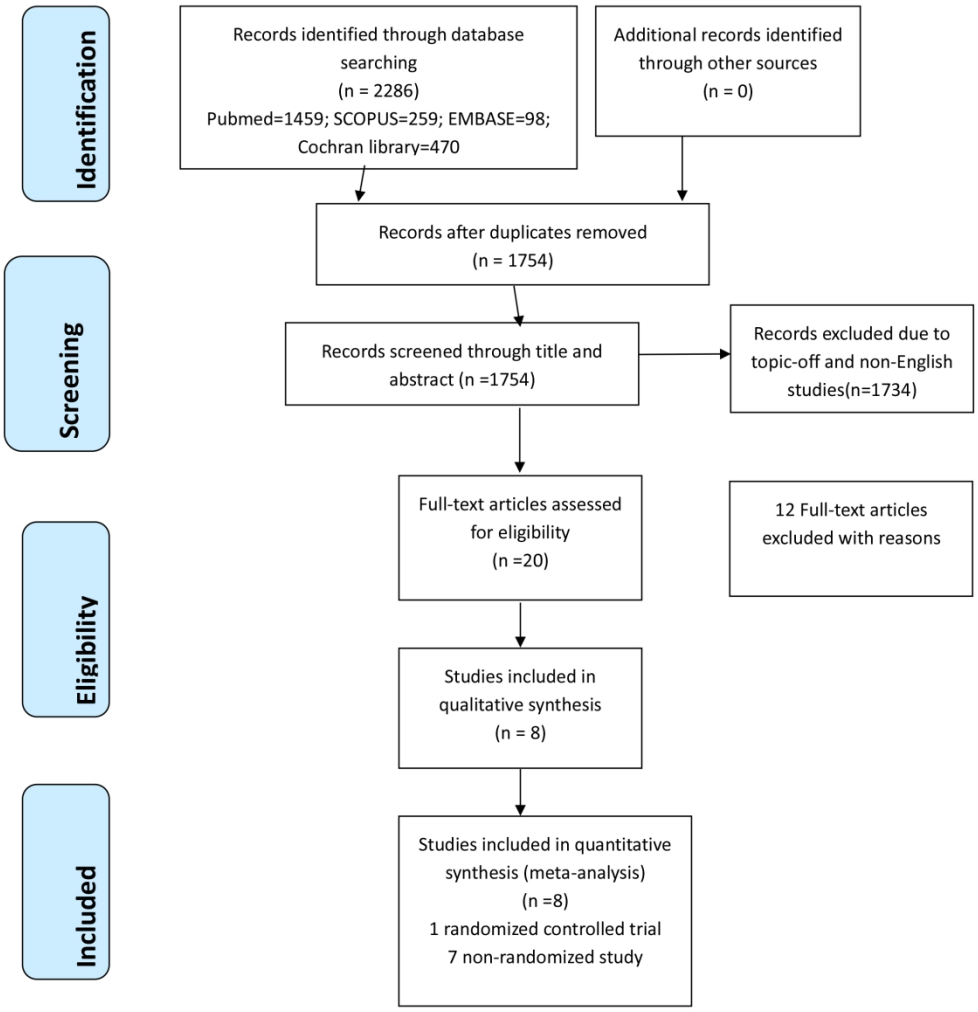


Figure 1. Study flow diagram

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Study flow diagram

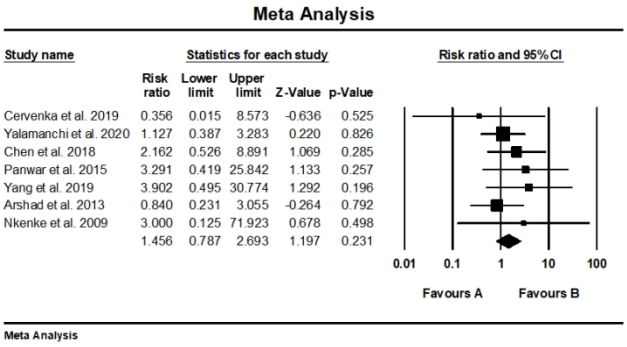


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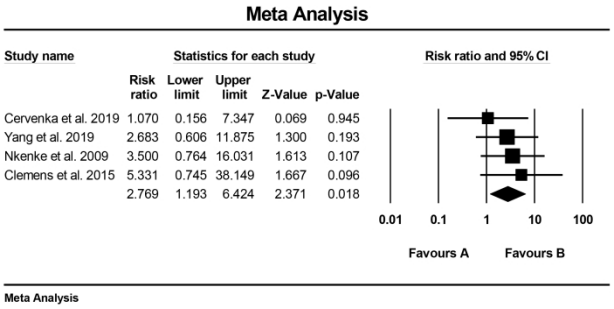


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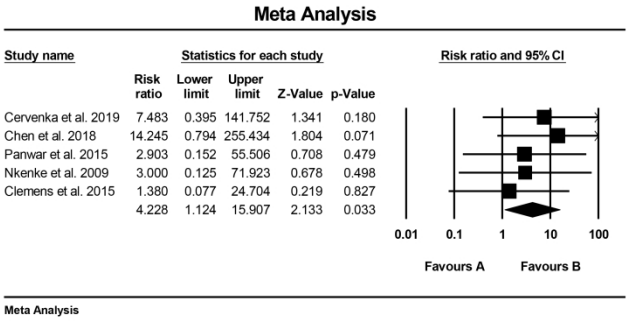


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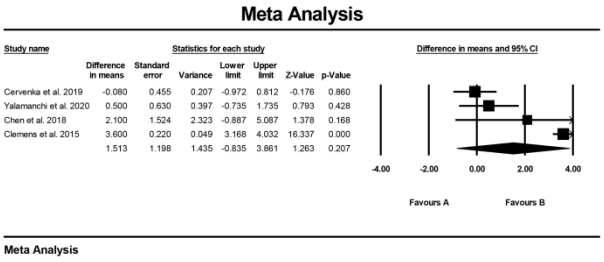


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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			1
Title	1	Identify the report as a systematic review.	1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	2
INTRODUCTION			3,4
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	3,4
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	4
METHODS			4-7
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	5,6
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	5
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	5
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	6
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	6
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	5,6,7
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	5
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	6,7
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	7
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	5,7
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	7
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	7
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	7
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	7
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	7
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	7
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	-

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	7,8
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	8
Study characteristics	17	Cite each included study and present its characteristics.	8, Table 1
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	10
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	11-15
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	11-15
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	11-15
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	-
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	-
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	-
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	-
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	15
	23b	Discuss any limitations of the evidence included in the review.	17,18
	23c	Discuss any limitations of the review processes used.	17,18
	23d	Discuss implications of the results for practice, policy, and future research.	18
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	4
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	4
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	4
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	19
Competing interests	26	Declare any competing interests of review authors.	19
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	-

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Table S1: Search terms

	Search terms			No.
Pubmed 29-04-2021	(((((((head and neck neoplasms[MeSH Terms]) OR (microsurgical free flap[MeSH Terms])) AND (nursing, perioperative[MeSH Terms])) OR (specialized care unit)) AND (ward care)) OR (care, surgical intensive[MeSH Terms])) AND (postoperative complications[Title/Abstract])) OR (flap failure[Title/Abstract])			1459
SCOPUS 29-04-2021	head and neck tumor OR free flap reconstruction AND intensive care unit OR surgical intensive care unit AND intensive care unit OR surgical intensive care unit OR specialized care AND graft failure OR postoperative complications			259
EMBASE 29-04-2021	('head and neck tumor'/exp OR 'free flap reconstruction'/exp) AND ('intensive care unit'/exp OR 'surgical intensive care unit'/exp) AND ('intensive care unit'/exp OR 'surgical intensive care unit'/exp OR 'specialized care'/exp) AND ('graft failure'/exp OR 'postoperative complications'/exp)			98
Cochran Library 29-04-2021	#1	MeSH descriptor: [Head and Neck Neoplasms] explode all trees		470
	#2	specialized care unit		
	#3	MeSH descriptor: [Free Tissue Flaps] explode all trees		
	#4	MeSH descriptor: [Intensive Care Units] explode all trees		
	#5	MeSH descriptor: [Postoperative Complications] explode all trees 39311		
	#6	flap loss		
	#7	#1 OR #3 AND #2 AND #4 AND #6		
	#8	#7 AND #5		

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Table S2 Excluded studies and reason for exclusion

	Study	Reason for exclusion
1	Godden et al. 1999 [1]	Included patients who underwent tumor resection without flap reconstruction
2	Moreno and Bonilla-Velez[2]	Compared three different protocols. Two protocol admitted to the ICU for overnight before transfer to the intermediate care and to the general floor at physician discretion.
3	EDWARD et al. 2002 [3]	Although the author stated that some patients were unadmitted to the ICU, but no outcome were reported regarding patients in Non-ICU setting
4	Vimawala et al[4]	Conference poster
5	Morse et al 2019 [5]	All groups admitted to the ICU before transfer to the Non-ICU setting.
6	Yu et al. 2019 [6]	All patients were transferred to the intermediate level care.
7	Mathew et al 2010[7]	Only 7 patients were transferred to the ICU
8	Dautremont et al 2013[8]	Patients in both groups were admitted to the ICU.
9	Bhama et al 2013[9]	Patients were admitted to open or close ICU.
10	Patel et al. 2017 [10]	No clear information about the three-protocols used (ICU, intermediate level, and surgical ward).
11	Al-Dam et al. 2014[11]	No clear information about the two-protocol used (ICU, intermediate level). Flap failure or other outcomes of interest were not reported in both groups.
12	Bertelsen et al. 2020 [12]	Two different protocols were used but, both group were admitted to the ICU for overnight before transfer to the ward.

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